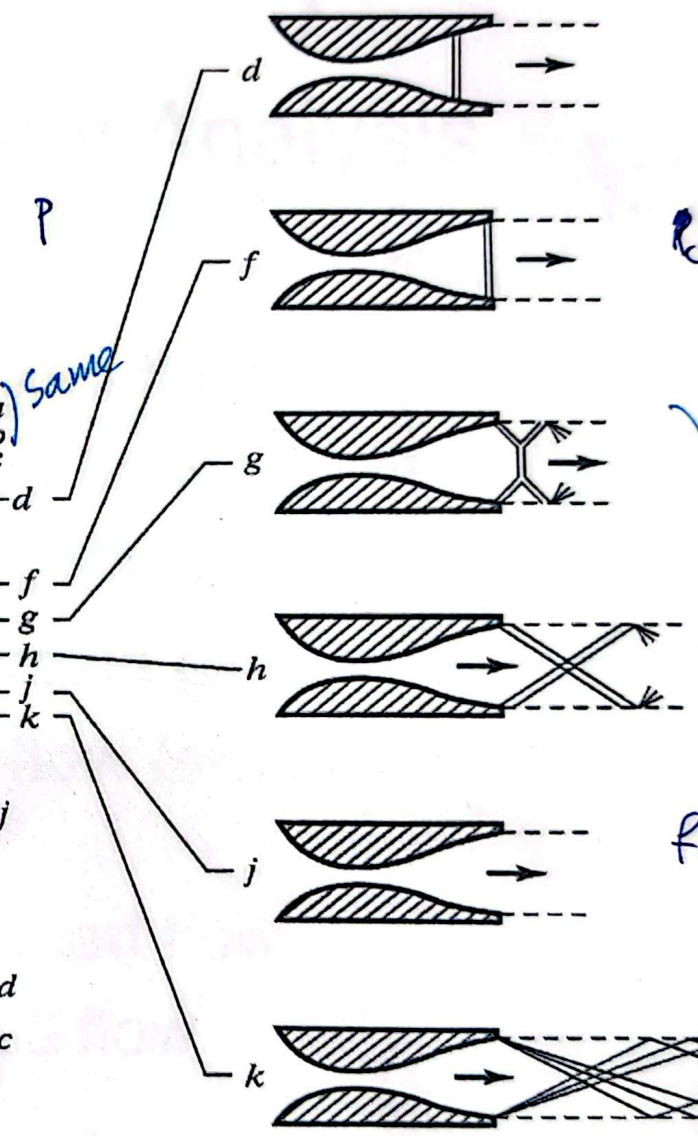
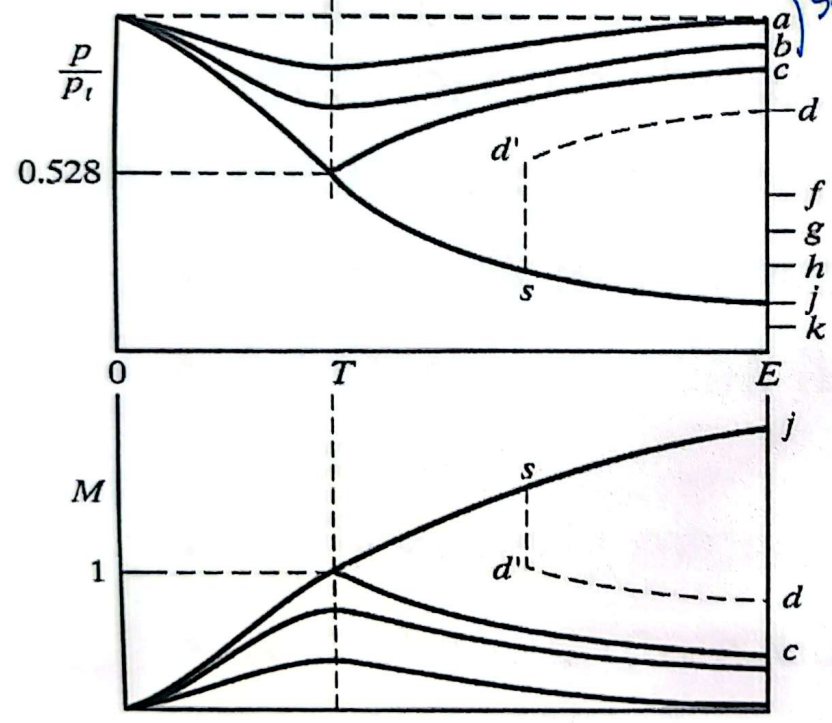
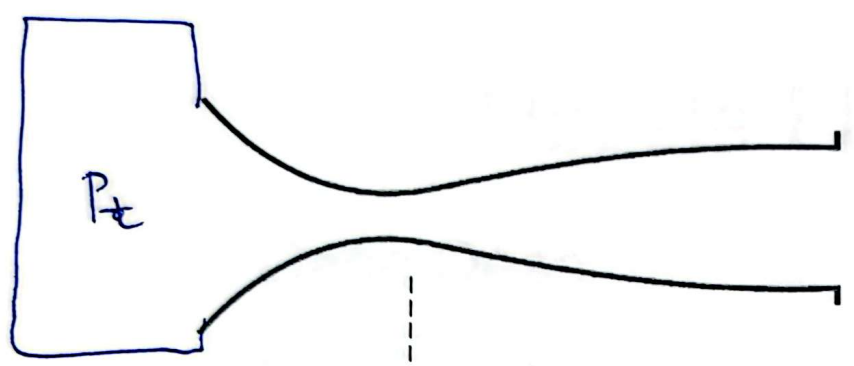




Critical

over-expanded
same

fully
expanded

under-expanded

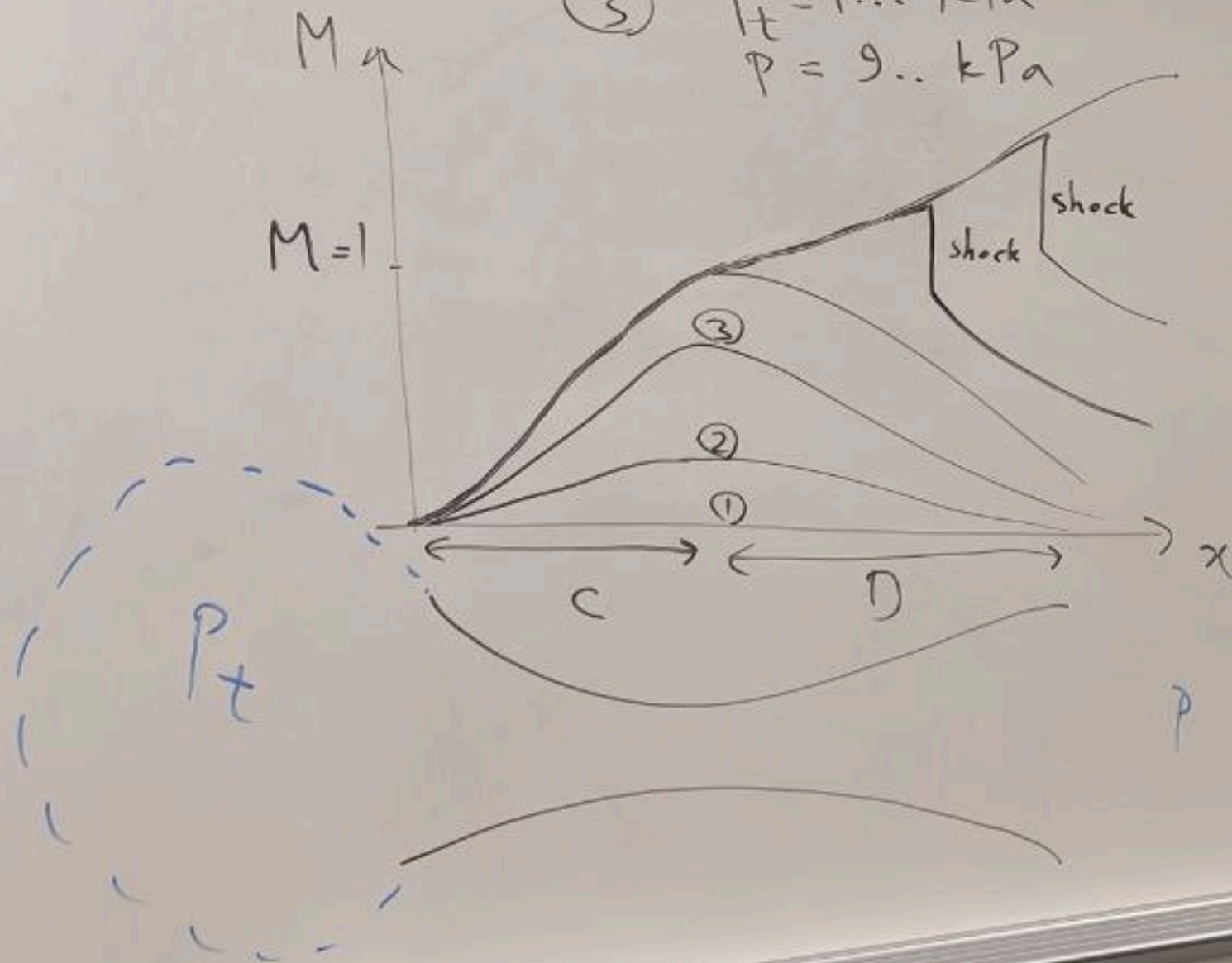
Simplify to 7 conditions

$P_t = 1000 \text{ kPa}$, P changes

① $P_t = P \rightarrow$ No flow

② $P_t = 1000 \text{ kPa}$, $P = 999 \text{ kPa}$

③ $P_t = 1000 \text{ kPa}$
 $P = 900 \text{ kPa}$



④ $P_t = 1000 \text{ kPa}$

P is such that
 P at throat equals

⑤ $P_t = 1000 \text{ kPa}$
 $P < P_4$

⑥ $P_t = 1000 \text{ kPa}$
 $P < P_5$

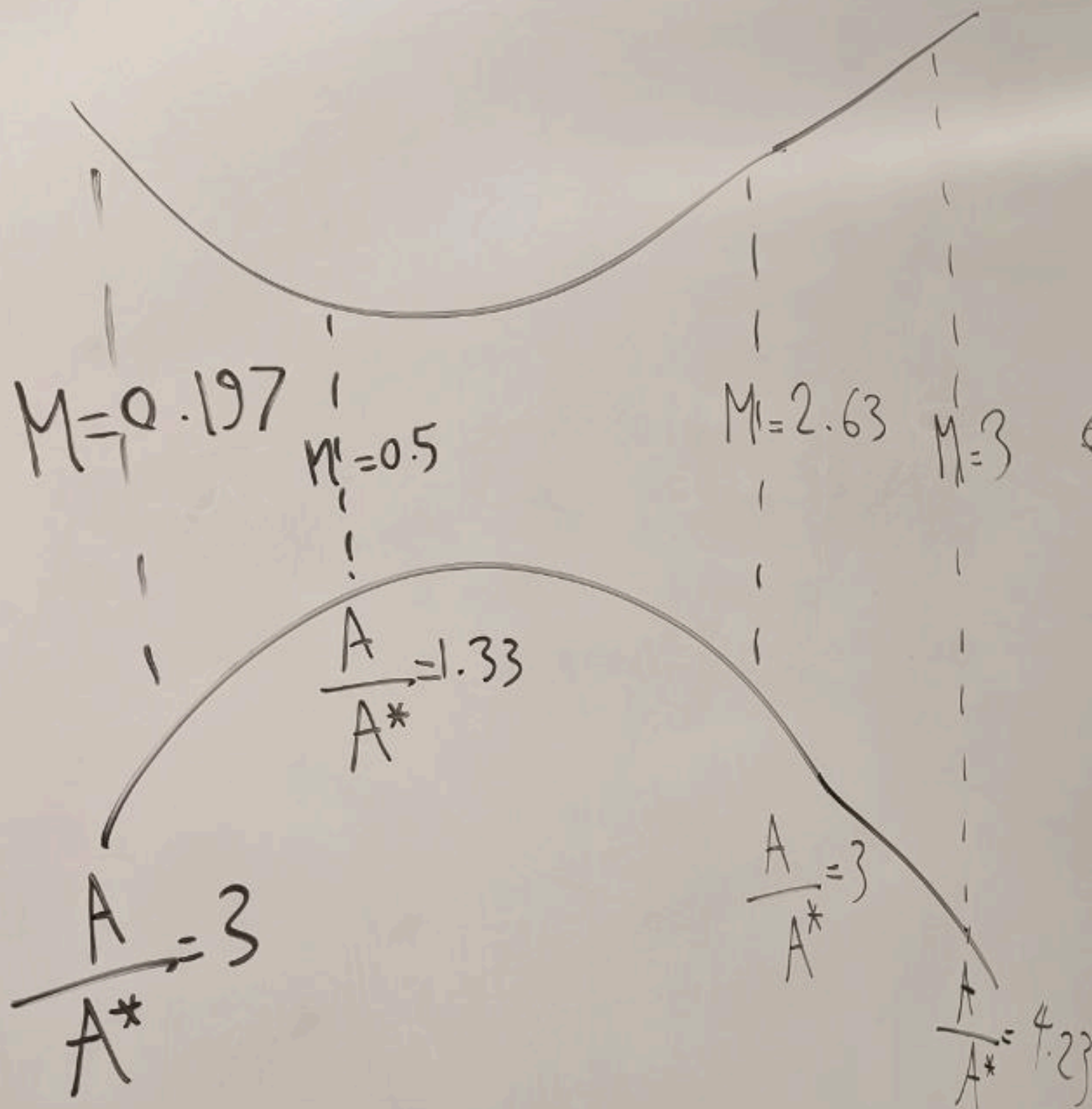
$$\frac{P}{P_t} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}}, \text{ put } \begin{cases} M=1 \\ \gamma=1.4 \end{cases}$$

get $\downarrow$
 $0.528 P_t \rightarrow M_{th} = 1 = M^*$

⑦ $\frac{P_t}{P} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}}$

My nozzle has for example
 $\frac{A}{A^*} = 5$, online Calc $M_{sup} = 3.17$

$\frac{P}{P_t} = 0.02$ put this ratio



$$\frac{A}{A^*}=3 \rightarrow M_{\text{sup}}=2.63$$

$\Leftarrow$

$$\frac{A}{A^*}=3 \rightarrow M_{\text{sub}}=0.197$$

$$\frac{A}{A^*}=3, M=M_{\text{sub}}$$

$$M = 0.197$$

$$M = 2.63 \leftarrow$$

$$\frac{A}{A^*} = 3$$

$$\frac{A}{A^*} = 3$$

$$\frac{A}{A^*} = 3 \rightarrow M_{sup} = 2.63$$

$$\frac{A}{A^*} = 3 \rightarrow M_{sub} = 0.197$$

$$\frac{A}{A^*} = 3, M = M_{sup}$$

$$\frac{A}{A^*} = 3, M = M_{sup}$$

$$\frac{A}{A^*} = 2 \xrightarrow[\text{calc}]{\text{online}}$$

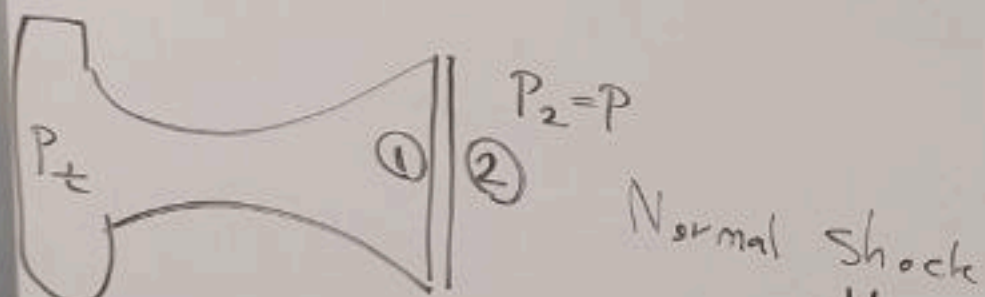
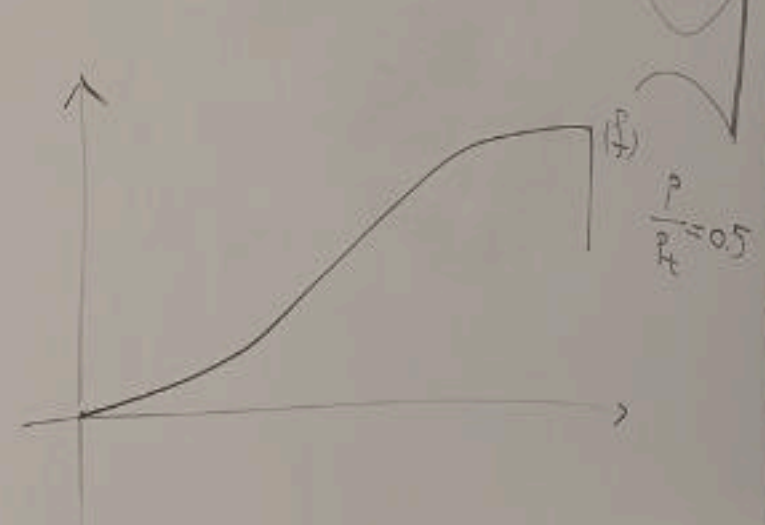
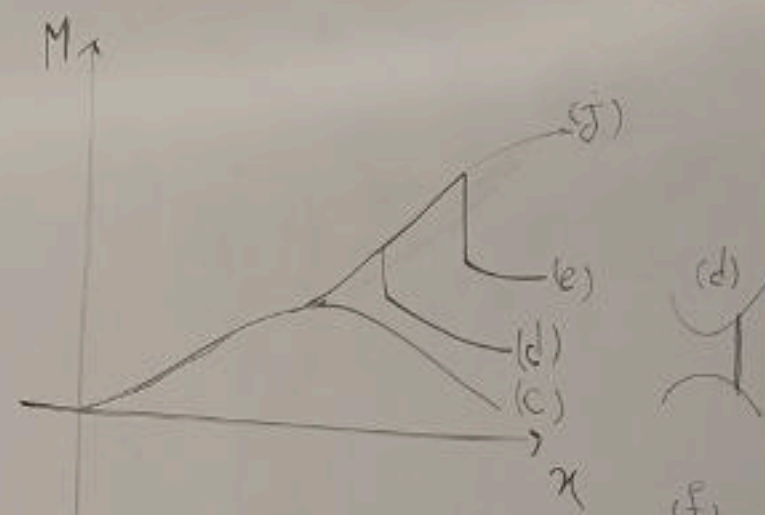
$$\left(\frac{P}{P_t}\right)_{\text{sup}} = 0.09 \quad (\text{case j})$$

$$\left(\frac{P}{P_t}\right)_{\text{sub}} = 0.937 \quad (\text{case c})$$

$$\left(\frac{P}{P_t}\right) = 0.8 \quad (// d)$$

$$0.6 \quad (// e)$$

$$N.5 \text{ at exit, } 0.5 \quad (// f)$$



$$\frac{A}{A^*} = 2 \xrightarrow{O.C.} M_1 = 2.197 \xrightarrow{P_2/P_1 = 5.46}$$

$$\frac{P_2}{P_t} = \frac{P}{P_t} = \frac{P_2}{P_1} \times \frac{P_1}{P_t} = 0.5$$

$$\left\{ \begin{array}{l} \frac{P_1}{P_t} = 0.093 \\ \text{Isentropic} \end{array} \right.$$

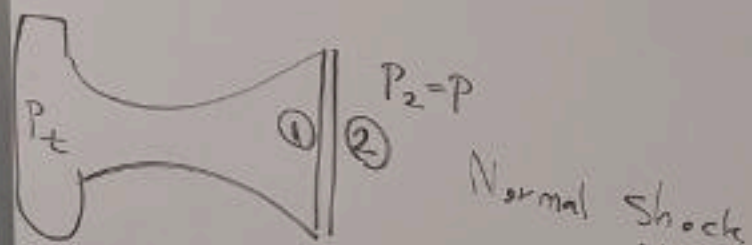
$$\frac{A}{A^*} = 2 \xrightarrow[\text{calc}]{\text{online}} \left(\frac{P}{P_t} \right)_{\text{sup}} = 0.09 \text{ (case j)}$$

$$\text{"} \quad \left(\frac{P}{P_t} \right)_{\text{sub}} = 0.937 \text{ (case c)}$$

$$\left(\frac{P}{P_t} \right) = 0.8 \quad (\text{" d})$$

$$0.6 \quad (\text{" e})$$

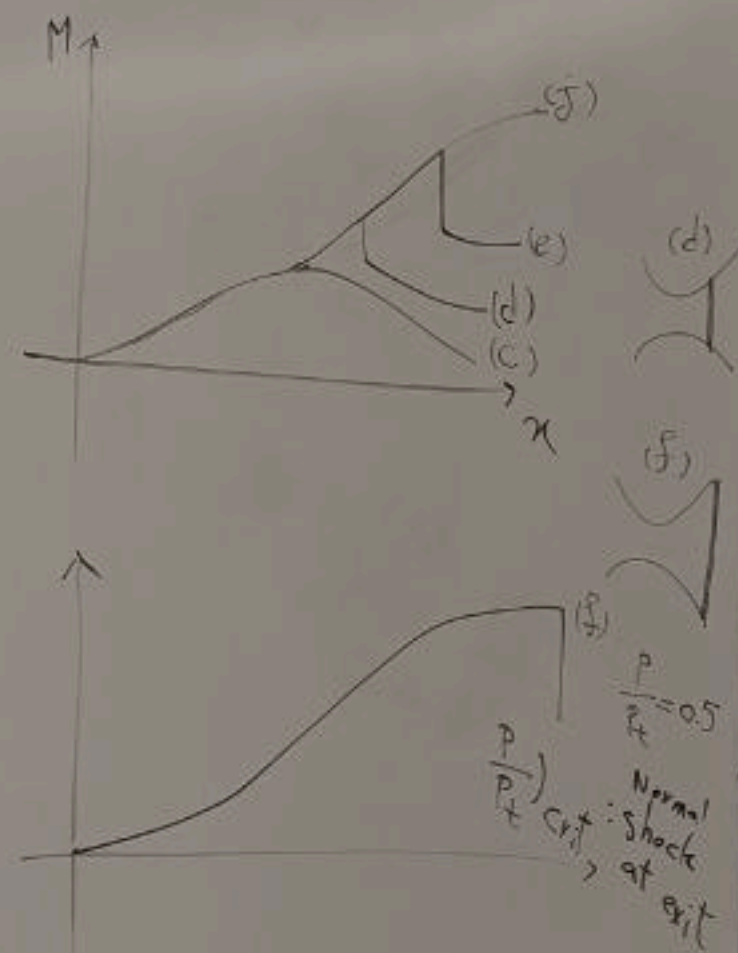
$$N.S \text{ at exit, } 0.5 \quad (\text{" f})$$



$$\frac{A}{A^*} = 2 \xrightarrow{O.C} M_1 = 2.177 \rightarrow \frac{P_2}{P_1} = 5.48$$

$$\frac{P_2}{P_t} = \frac{P}{P_t} = \frac{P_2}{P_1} \times \frac{P_1}{P_t} = 0.5$$

$$\left\{ \begin{array}{l} \frac{P_1}{P_t} = 0.093 \\ \text{Isentropic} \end{array} \right.$$



$$\frac{P}{P_t} = 0.4$$

$$\text{if } \frac{P_1}{P_t} < \frac{P}{P_t} < \frac{P_2}{P_t}$$

We have shock oblique for our case

$$0.09 < \frac{P}{P_t} < 0.5$$

We have oblique shock

$$\frac{P}{P_t} = 0.4$$



Note $\frac{P}{P_t} < \frac{P}{P_t}_{crit} < \frac{P}{P_t}_{sub}$

If $\frac{P}{P_t} < \frac{P}{P_t}_{sup}$

We have Normal shock in nozzle

For our case

$$0.5 < \frac{P}{P_t} < 0.937$$



Expansion wave at exit

Under-expanded

$\frac{P}{P_t} = 0.09 \rightarrow$ fully supersonic expanded
 $= \frac{P}{P_t}_{sup}$

If $\frac{P}{P_t} < \frac{P}{P_t}_{sup} < \frac{P}{P_t}_{crit}$ "over-expanded"

We have shock at outside oblique

For our case in

$$0.09 < \frac{P}{P_t} < 0.5$$

We have oblique shock



(f)

$\frac{P}{P_t} = 0.5$

Normal shock at exit

Nozzle Flow Analysis

Example

$$\frac{P}{P_t} \frac{A}{A^*} M_{th} = 2$$

$$\frac{P}{P_t} > 0.937$$

Case Descriptions:

- (i) Subsonic unchoked flow
- (ii) Subsonic choked flow
- (iii) Shock in the nozzle
- (iv) Shock at the nozzle exit
- (v) Overexpanded flow (shocks outside nozzle)
- (vi) Perfectly expanded flow
- (vii) Underexpanded flow

$$\frac{P}{P_t} = 0.937$$

$$P_b = P_e$$

$$\left(\frac{P}{P_t}\right)_{crit} < \frac{P}{P_t} < \left(\frac{P}{P_t}\right)_{sub}$$

$$\frac{P}{P_t} = \left(\frac{P}{P_t}\right)_{crit}$$

$$P_b > P_e$$

$$\left(\frac{P}{P_t}\right)_{sup} < \frac{P}{P_t} < \left(\frac{P}{P_t}\right)_{crit}$$

$$P_b = P_e \rightarrow$$

$$\frac{P}{P_t} = \left(\frac{P}{P_t}\right)_{sup}$$

$$P_b < P_e$$

$$\frac{P}{P_t} < \left(\frac{P}{P_t}\right)_{sup}$$

$$M_{th} = 1$$

same flow inside nozzle

expansion fan outside

P_e P or P_b

If flow is fully expanded $P_e = P_b$

If flow is under- " or over-expanded

$P_e \neq P_b$

In both ① and ②

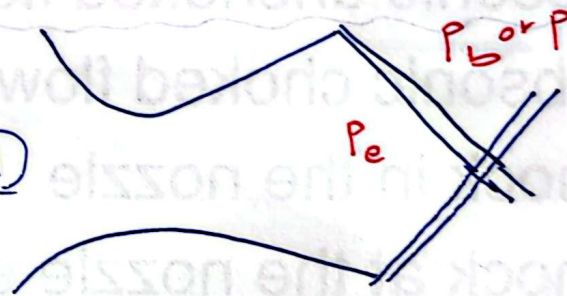
P_e is computed from

$$\frac{P_e}{P_t} = \left(1 + \frac{\gamma-1}{2} M_{sup}^2\right)^{\frac{\gamma}{\gamma-1}}$$

M_{sup} is read from

$$\left(\frac{A}{A^*}\right)_{sup}$$

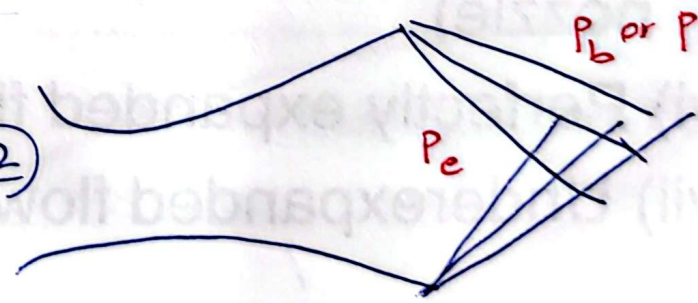
①



$P_e < P_b$

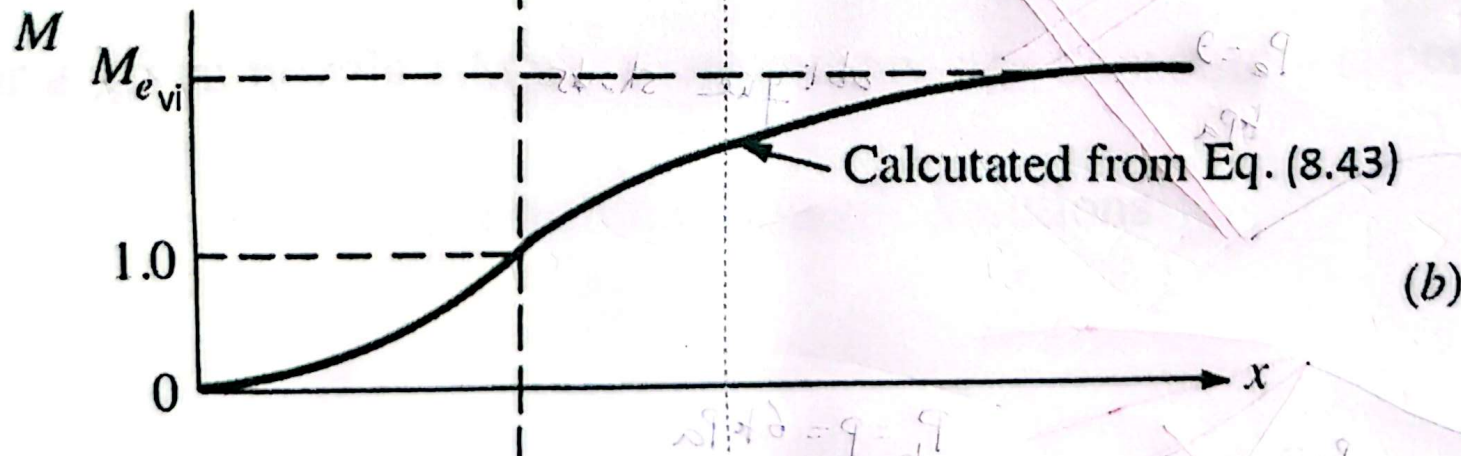
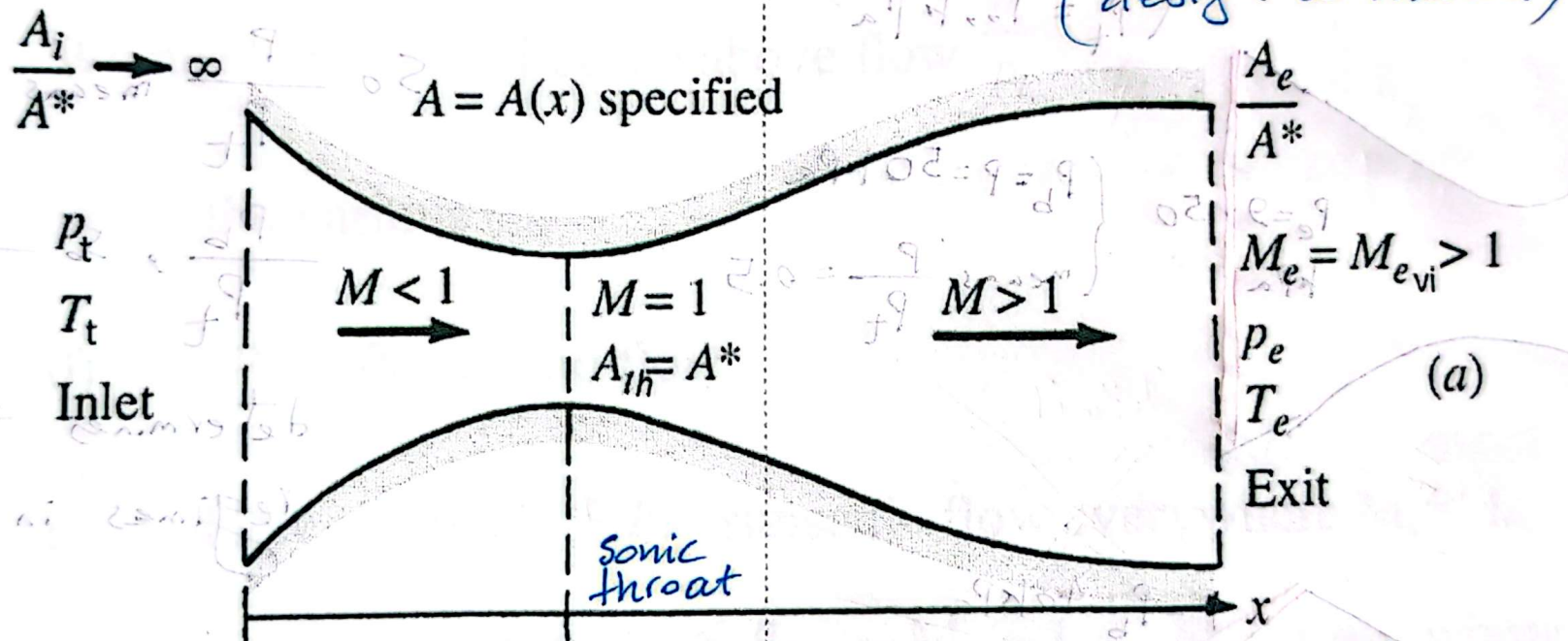
So a shock occurs to match and increase P_e to P_b

②

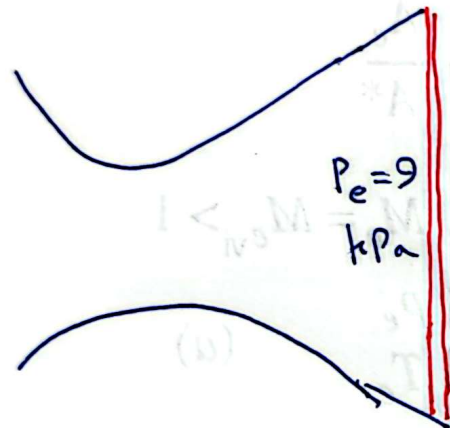


$P_e > P_b$

So an expansion wave occurs to reduce pressure



Ex) $\frac{A}{A^*} = 2 \xrightarrow[\text{cal}]{\text{online}}$ $\begin{cases} \frac{P}{P_t} = 0.09 \\ P_t = 1.0 \text{ kPa} \end{cases} \Rightarrow P = 9 \text{ kPa} = P_e$

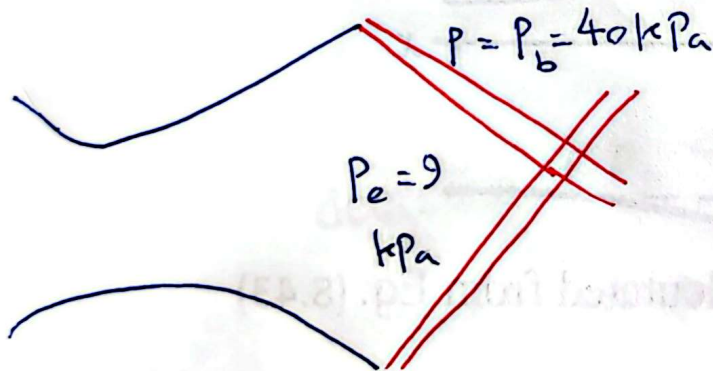


50 $\begin{cases} P_b = P = 50 \text{ kPa} \\ \text{means } \frac{P}{P_t} = 0.5 \end{cases}$

So $\frac{P}{P_t}$ means

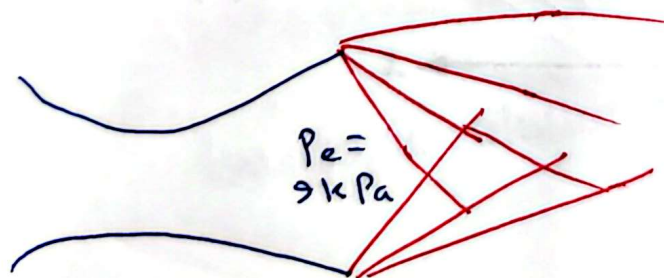
$\frac{P_b}{P_t}$, so $\frac{P_b}{P_t}$

determines flow regimes in nozzle



$P = P_b = 40 \text{ kPa}$
 $P_e = 9 \text{ kPa}$

oblique shock



$P_b = P = 6 \text{ kPa}$

Expansion waves

We need $p_e < p_t$ For the above flow $\frac{p_e}{p_t} = \frac{P_{e\ vi}}{p_t}$

(numbering in page 13)

Review the various cases:

0) $p_e = p_t \Rightarrow$ no flow

i) decrease of $p_{e_i} < p_t$ subsonic flow everywhere $M_t < 1$ (unchoked)

ii) further decrease $p_{e_{ii}} < p_{e_i}$ $M_t = 1$, $M < 1$ elsewhere (choked)

For a given nozzle ($A(x)$) there is only one subsonic – supersonic

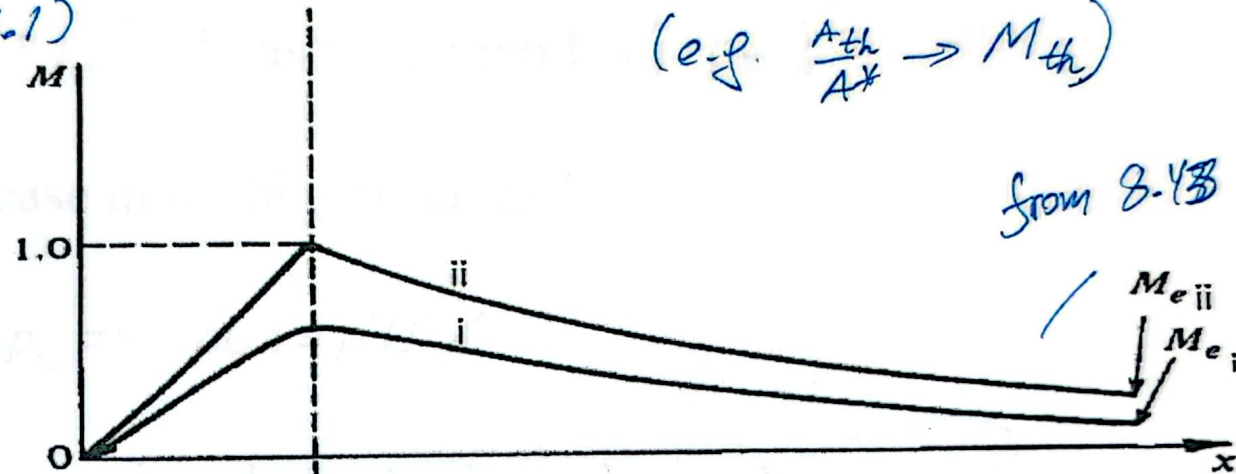
isentropic solution but infinite subsonic solutions

(vi)

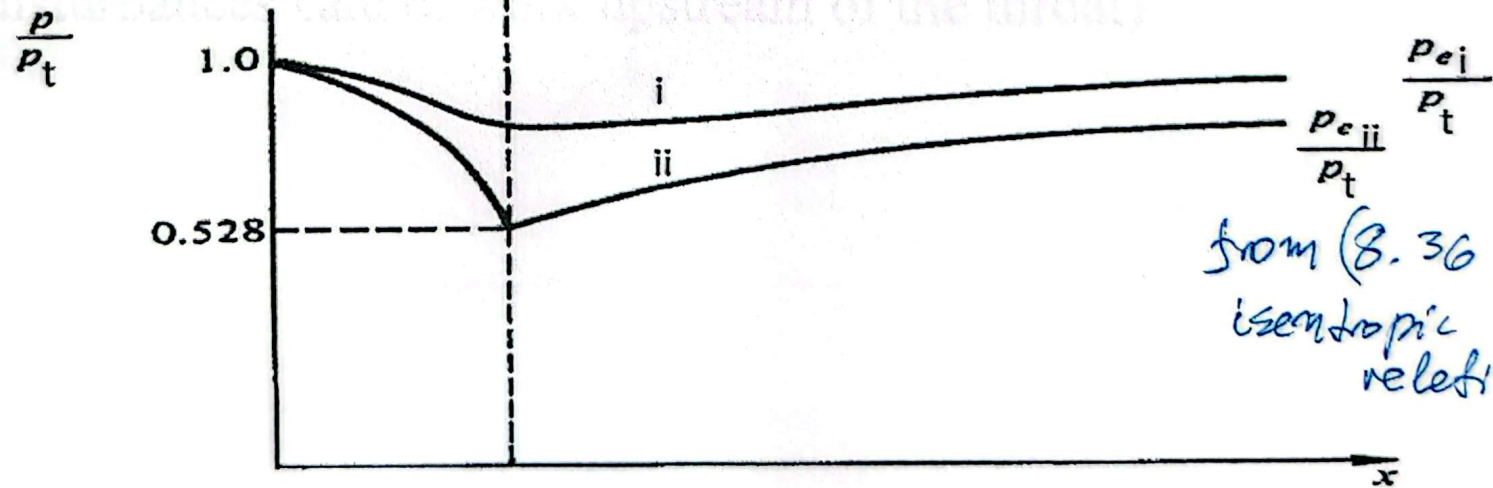
(i)

(i) $\frac{p_e}{p_t} \rightsquigarrow M_e \rightsquigarrow \frac{A_e}{A^*} \rightsquigarrow A^* < A_{th}$ from $\frac{A}{A^*} \rightsquigarrow M$
 (table 8.1) (e.g. $\frac{A_{th}}{A^*} \rightarrow M_{th}$)

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from 8.43 $\frac{A^*}{A} = f(M)$
 $\rightarrow M = \dots$



from (8.36)
 isentropic
 relation

same for p, T

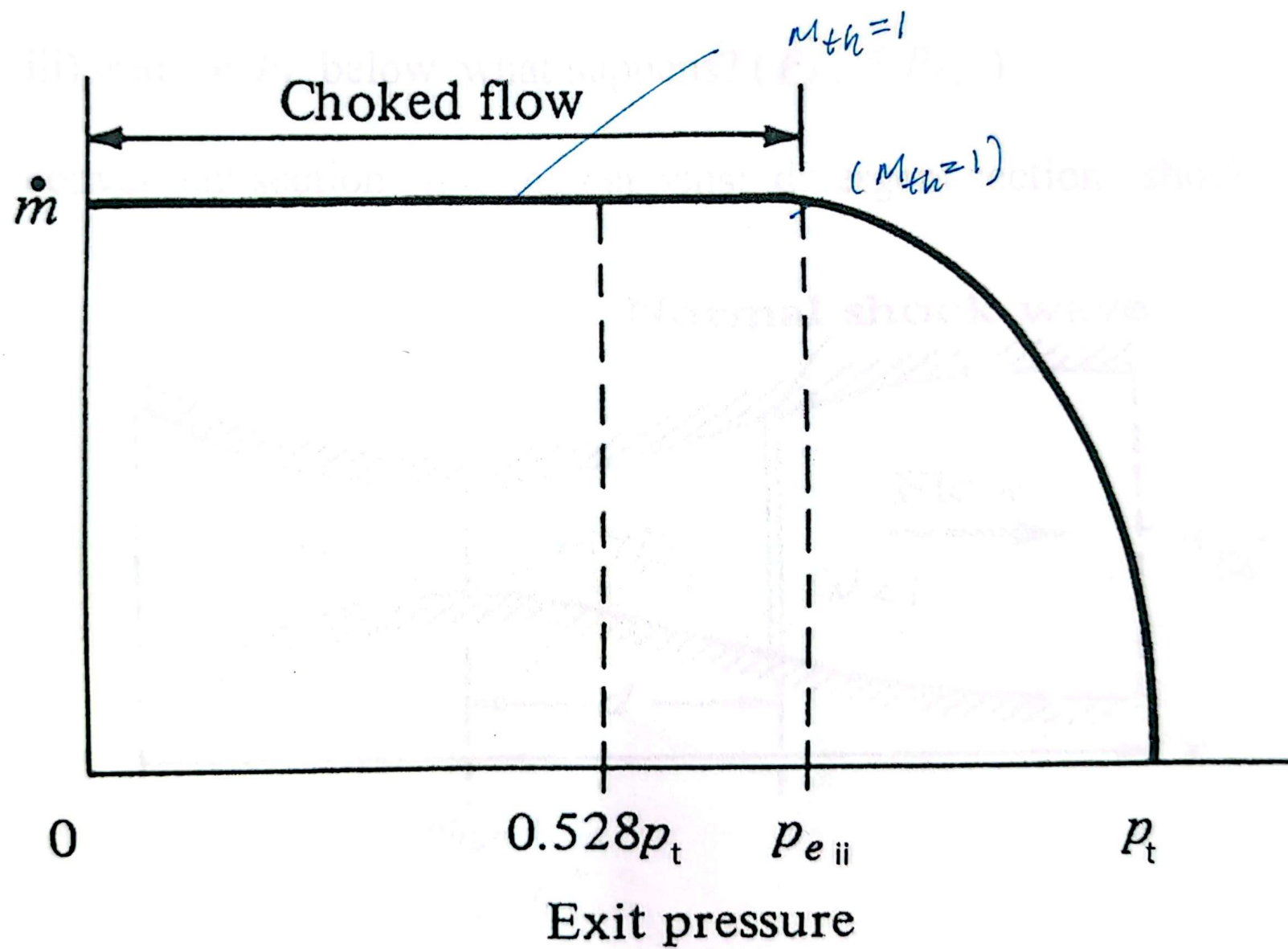
Consider $\dot{m} = \overbrace{\rho_{th} U_{th} A_{th}}^{\text{throat}}$ as $p_e \downarrow$ then $U_{th} \uparrow$ $\rho_{th} \downarrow$

but the increase in U_{th} bigger $\Rightarrow \dot{m} \uparrow$

when $p_e = p_{eII} \Rightarrow \rho U A = \rho^* U^* A^*$

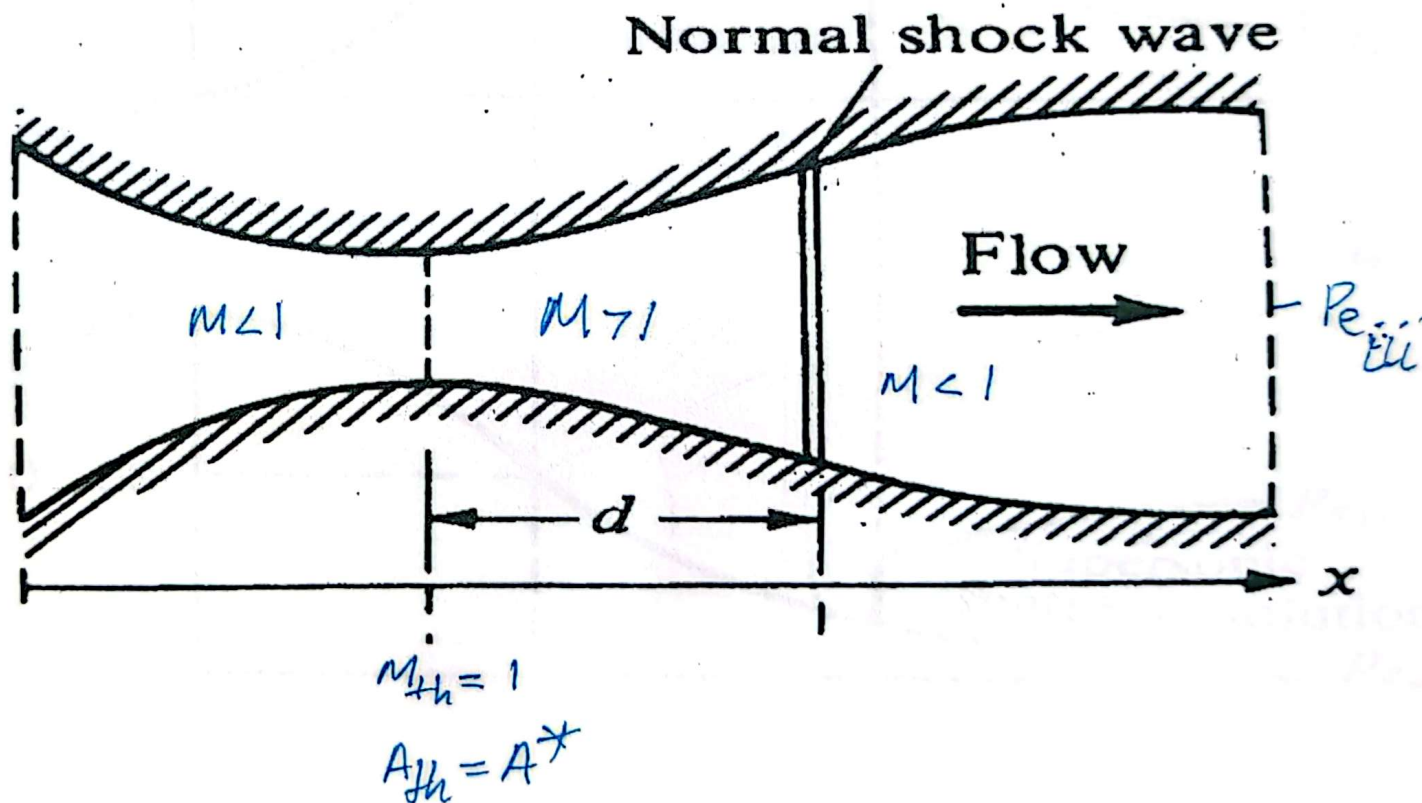
if p_e is further reduced $\Rightarrow \dot{m}$ is unchanged: choked flow

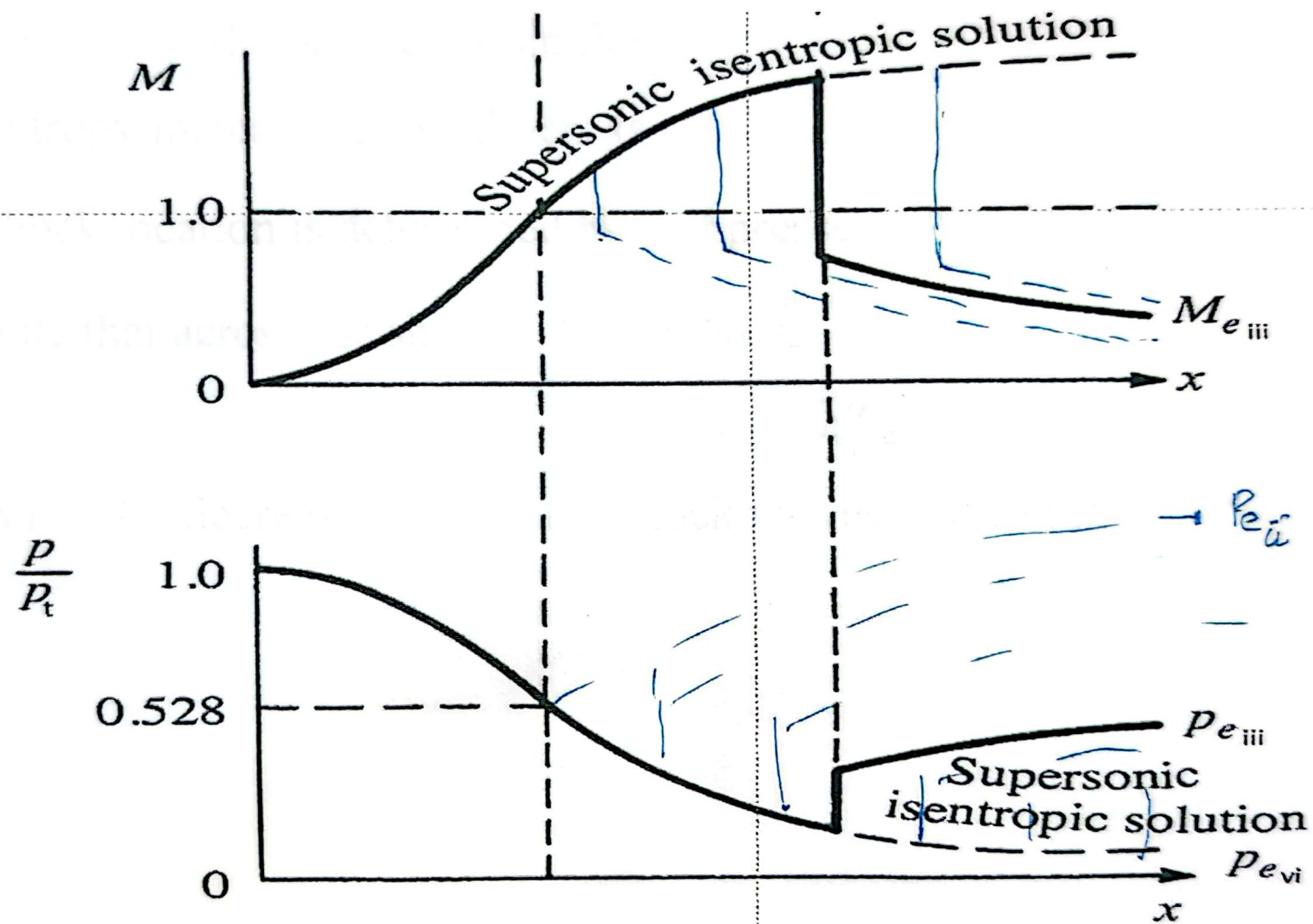
(disturbances cannot work upstream of the throat)



iii) reduce P_e below what happens? ($P_{e_{III}} < P_{e_{II}}$)

convergent section: nothing happens; divergent section: shock





behind the shock: isentropic flow

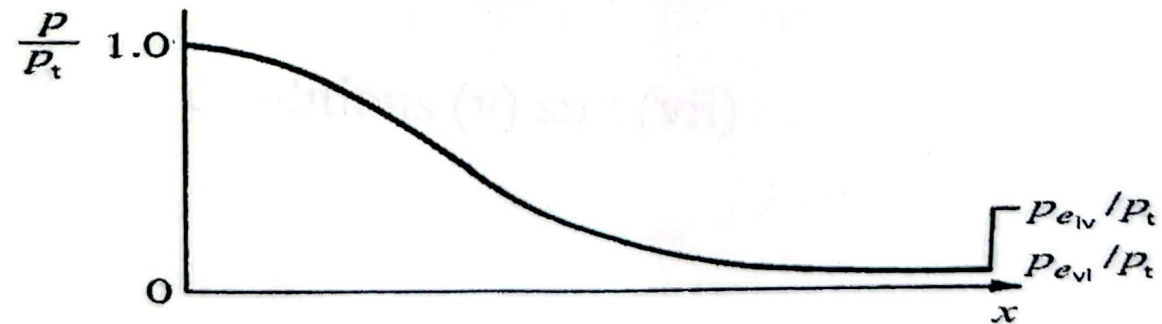
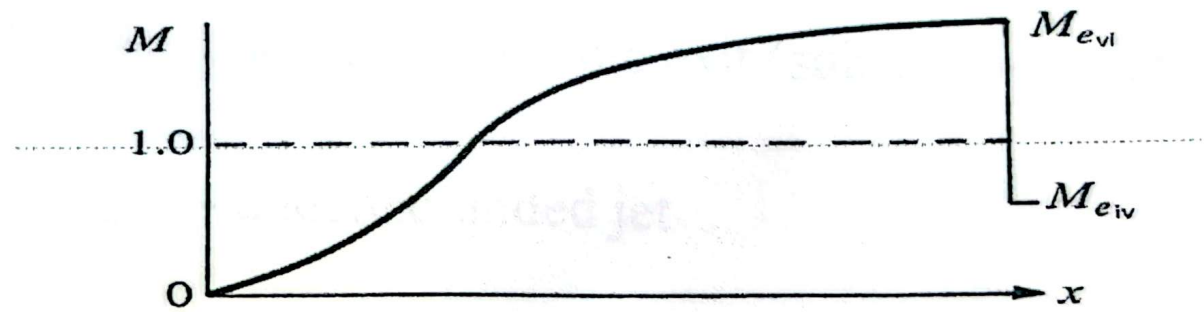
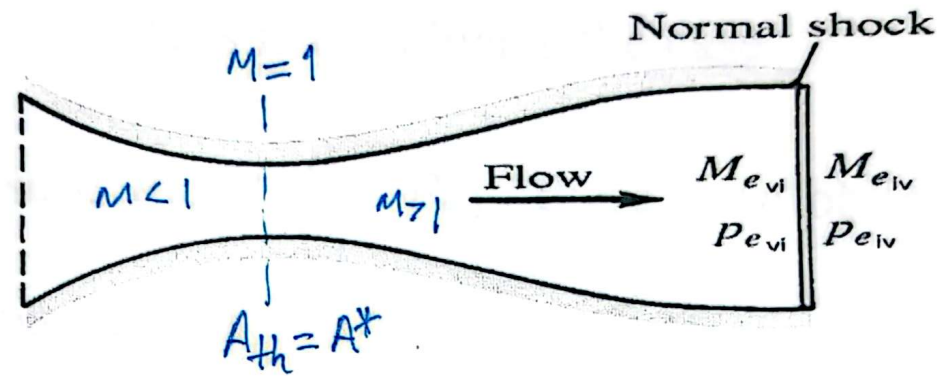
entropy increase across shock ($s_2 > s_1$)

shock location is determined by exit pressure $p_{e_{III}}$ (see later)

note that across the shock: $p_{t1} A_1^* = p_{t2} A_2^*$

$$(p_{t1} > p_{t2} \Rightarrow A_1^* < A_2^*)$$

iv) as p_e decreases to $p_{e_{IV}}$ the shock moves at the exit



Up to now $p_b = p_e$ This is not always the case

v) For back pressure $p_{e_{iv}} > p_b > p_{e_{vi}}$ there is an oblique shock at the exit – overexpanded jet

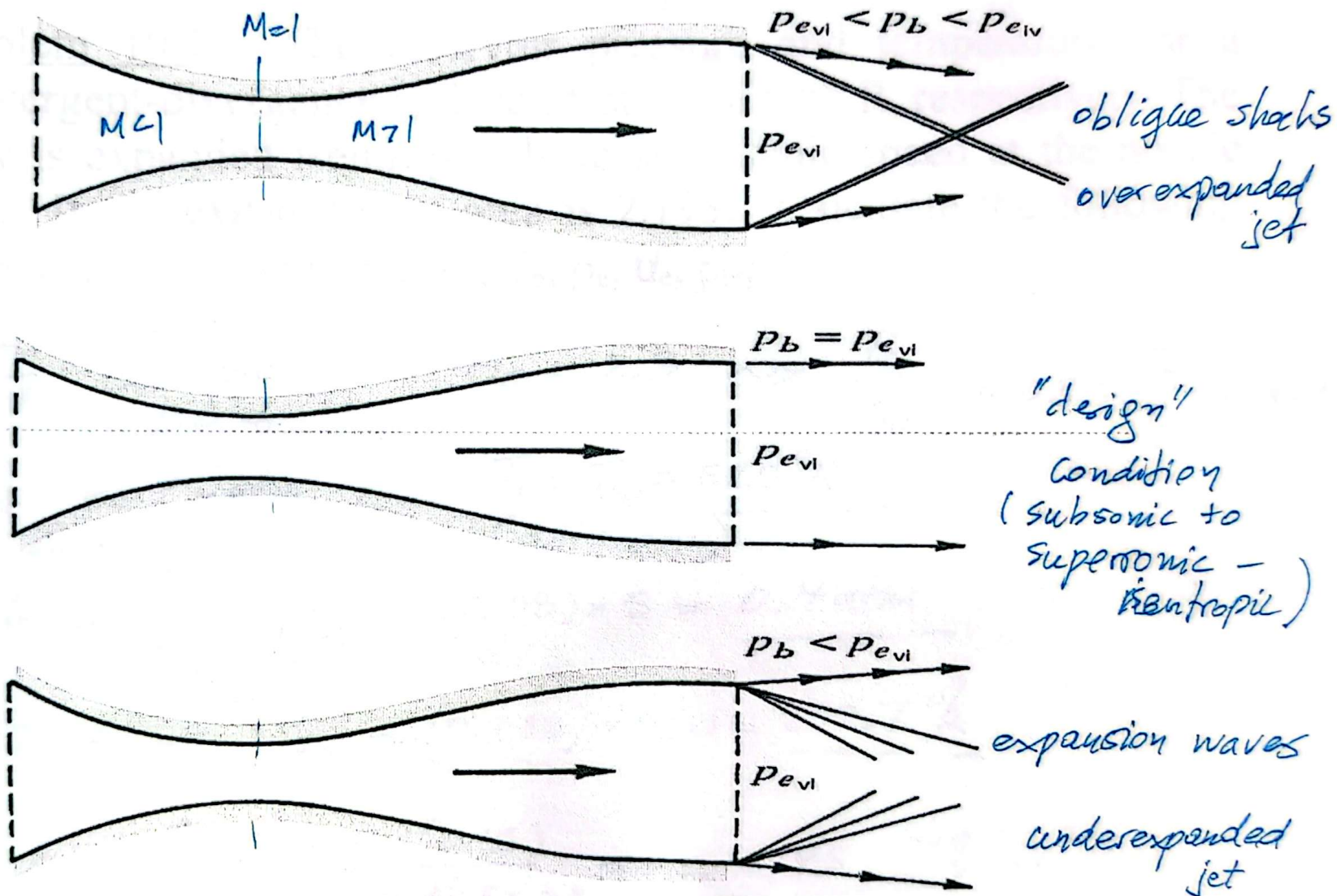
vi) if $p_b = p_{e_{vi}}$ normal isentropic exit (subsonic-supersonic)

vii) $p_b < p_{e_{vi}}$ - underexpanded jet

expansion waves after the nozzle (p_e can not go lower than $p_{e_{vi}}$)

note: $p_b \neq p_e$ for conditions (v) and (vii)

only



flow in the nozzle stays the same for conditions (v), (vi), (vii)

Problem 10.1: The reservoir pressure and temperature for a convergent-divergent nozzle are P_t and T_t 5 atm and 520° R, respectively. The flow is expanded isentropically to supersonic speed at the nozzle exit. If the exit-to-throat area is 2.193. Calculate the following properties at the exit: M_e , p_e , T_e , ρ_e , U_e , p_{te} , T_{te} .

$$\frac{A_e}{A^*} = 2.193 \xrightarrow[\text{tables}]{\text{isentropic}} M_e \approx 2.3 \xrightarrow{\quad} \frac{p_e}{p_{te}} = 0.08; \frac{T_e}{T_{te}} = 0.486$$

$$P_t = P_{te} = 5 \text{ atm} \\ (\text{isentropic})$$

$$T_t = T_{te} = 520^\circ \text{ R} \\ (\text{adiabatic})$$

$$p_e = \frac{p_e}{p_{te}} \cdot p_{te} = (0.08) \times 5 = \underline{0.4 \text{ atm}}$$

$$T_e = \frac{T_e}{T_{te}} \cdot T_{te} = (0.486) \times 520 = \underline{252.7^\circ \text{ R}}$$

$$\rho_e = \frac{p_e}{RT_e} = \frac{(0.4)(2116)}{(1716)(252.7)} = \underline{0.00195 \text{ slug/ft}^3}$$

$$U_e = M_e \underbrace{\sqrt{\gamma R T_e}}_{a_e} = 2.3 \times \underbrace{\sqrt{(1.4)(1716)(252.7)}}_{779.2 \text{ ft/sec}} = \underline{1792 \text{ ft/sec}}$$

Nozzle Flow Analysis

For a given area ratio: $\frac{A_e}{A_{th}}$, P_t , and P_b (or P_t/P_b)
 [$P_e = P_b$ except conditions v, vii]

- Determine P_t/P_b for subsonic choked flow (ii) $P_b = P_e$
 $(P_t/P_b)_{ii}$
 isentropic tables (using A_e/A^*) - subsonic solution

- Determine P_t/P_b for perfectly expanded flow (vi) $P_b = P_e$

- Determine P_t/P_b for shock at the exit (iv) $P_b = P_e$
 $(P_t/P_b)_{vi}$
 isentropic tables (using A_e/A^*) - supersonic solution

- Determine P_t/P_b for shock at the exit (iv)

$(P_t/P_b)_{iv}$

use M_{evi} for M_1 (Mach # before shock)
 normal shock tables: $M_2, \frac{P_2}{P_1}$
 $\rightarrow (P_e)_{iv}$

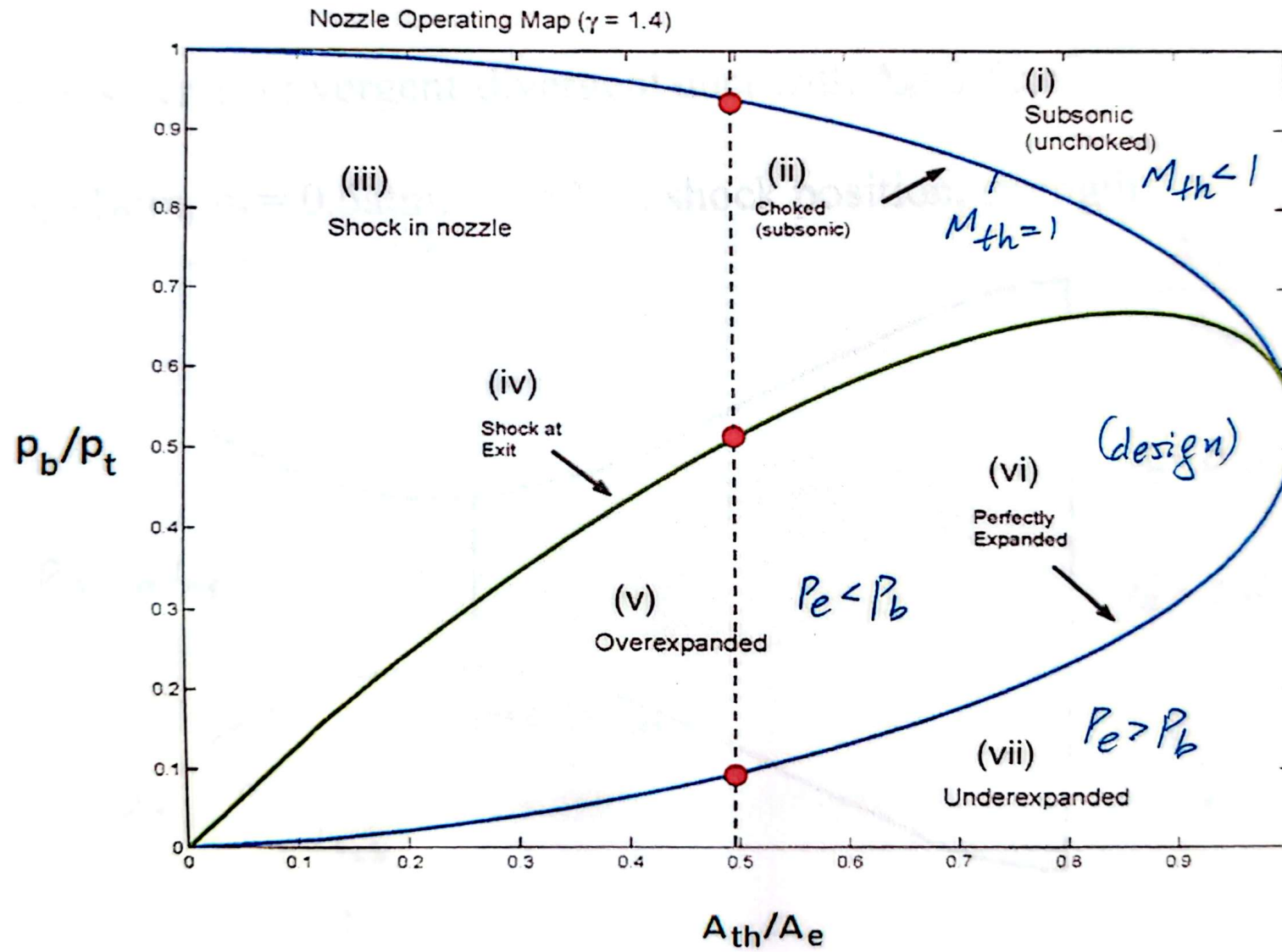
Example (with #s)

For a given area ratio (e.g. $A_e/A_{th} = 2.0$):

- Determine P_t/P_b for (ii) subsonic choked flow
 - Isentropic subsonic solution with $A_{th} = A^*$
subsonic with $A/A^* = 2.0$ gives $P_t/P = \underline{1.0670}$
- Determine P_t/P_b for (vi) perfectly expanded flow
 - Isentropic supersonic solution with $A_{th} = A^*$
supersonic with $A/A^* = 2.0$ gives $P_t/P = \underline{10.6459}$ and $M_e = 2.1972$
- Determine P_t/P_b for (iv) shock at nozzle exit
 - Set M_1 equal to Mach number from (vi), use shock jump relations
 $M_1 = 2.1972$ gives $P_2/P_1 = 5.4656$
 - Then $P_t/P_b = (P_t/P_1) \cdot (P_1/P_2) = \underline{10.6459/5.4656} = \underline{1.9478}$

shock at the exit

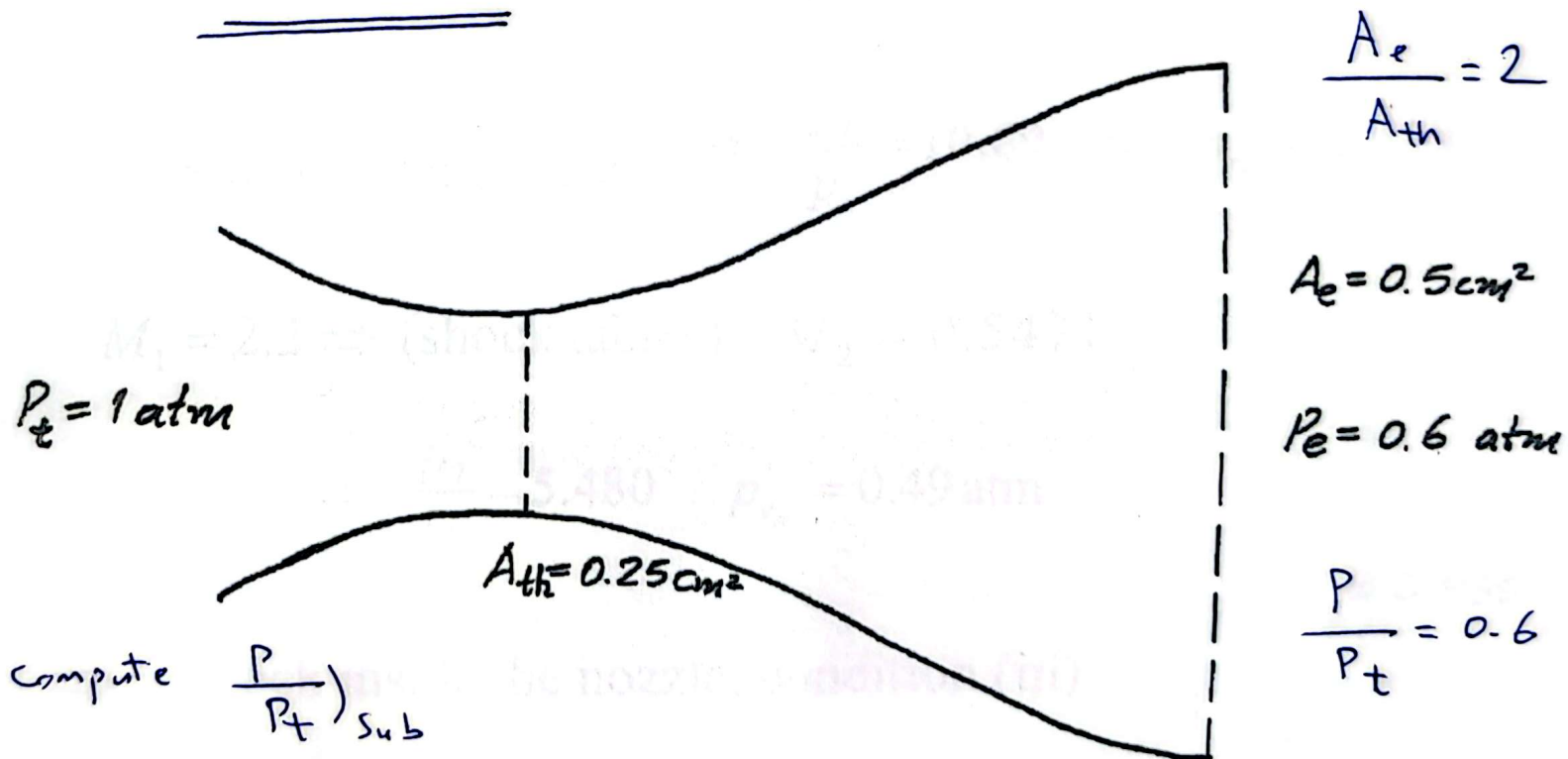
P_t



Example (finding normal shock location):

Consider a convergent-divergent duct with $A_e = 0.5 \text{ cm}^2$, $A_{th} = 0.25 \text{ cm}^2$

$p_t = 1 \text{ atm}$, $p_e = 0.6 \text{ atm}$; find M_e , shock position, strength



- ① compute $\left(\frac{p}{p_t}\right)_{sub}$
- ② compute $\left(\frac{p}{p_t}\right)_{crit}$

If $\left(\frac{p}{p_t}\right)_{crit} < \frac{p}{p_t} < \left(\frac{p}{p_t}\right)_{sub} \rightarrow$ there is shock in "nozzle"

Isentropic tables for $\frac{A_e}{A_t} = 2$:

$$\text{Subsonic: } M=0.31 \rightarrow \frac{p_t}{p_e} = 1.069 \Rightarrow p_{e_{ii}} = 0.935 \text{ atm}$$

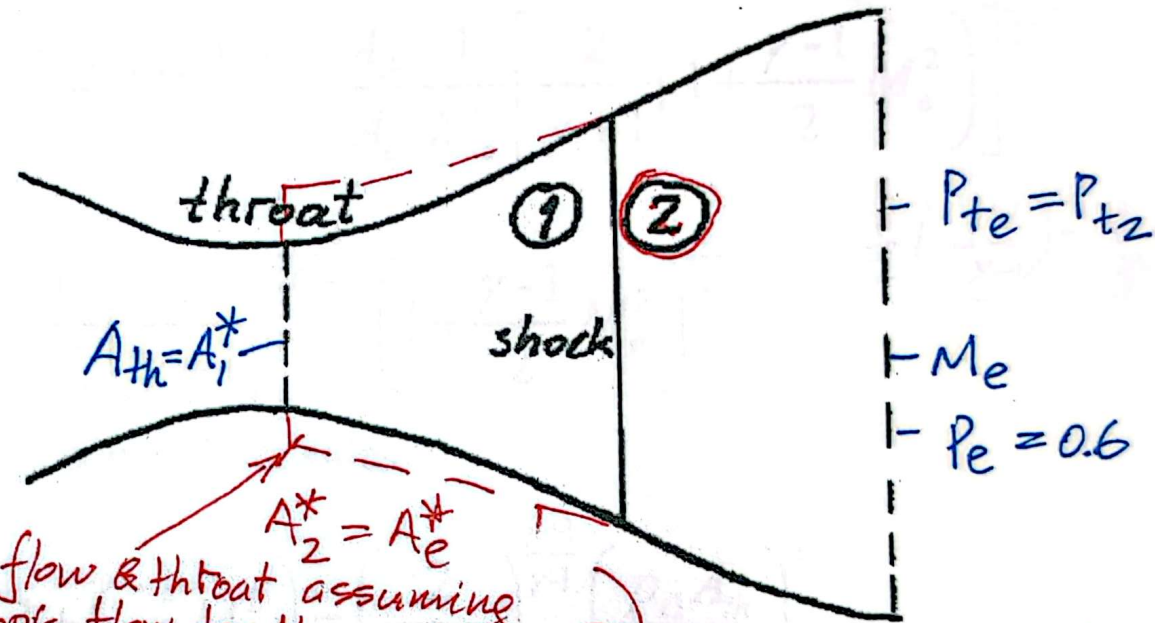
$$\text{Supersonic: } M=2.2 \rightarrow \frac{p_t}{p_e} = 10.69 \Rightarrow p_{e_{vi}} = 0.09 \text{ atm}$$

$$M_1 = 2.2 \Rightarrow (\text{shock tables}) \quad M_2 = 0.5471$$

$$\Rightarrow \frac{p_2}{p_1} = 5.480 \quad p_{e_{iv}} = 0.49 \text{ atm}$$

$\Rightarrow$ shock inside the nozzle, condition (iii)

$$\left(\begin{array}{l} p_{e_{ii}} = 0.935 \text{ atm} \\ p_{e_{iii}} = 0.6 \text{ atm} \\ p_{iv} = 0.49 \text{ atm} \end{array} \right)$$



(sonic flow & throat assuming isentropic flow for the conditions ②)

because m is constant (8.45)

$p_t A^*$ is constant throughout the nozzle so

$$\Rightarrow \frac{p_{te} p_e}{p_e p_{t1}} = \frac{A_e A_{th}}{A_e^* A_e}$$

throat

exit

$$p_{t1} A_t = p_{te} A_e^*$$

$$\frac{A_1^*}{p_{t2}} \parallel \frac{A_2^*}{p_{t2}}$$

$$A_2^* > A_1^*$$

$$\Rightarrow \frac{p_e}{p_{t1}} \left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{\frac{\gamma}{\gamma-1}} = \frac{A_{th}}{A_e} \frac{1}{M_e} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_e^2 \right) \right]^{\frac{(\gamma+1)}{2(\gamma-1)}}$$

$\swarrow$ p_{te}/p_e (8.36) A_e/A_e^* (8.42)

(algebra)

$$\Rightarrow \frac{p_e A_e}{p_{t1} A_{th}} = \frac{1}{M_e} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{\frac{1}{2}}$$

$\frac{1}{2} \left(\frac{\gamma+1}{\gamma-1} \right) - \frac{\gamma}{\gamma-1} = -\frac{1}{2}$

Cross-multiply and square both sides

$$M_e^2 \left(1 + \frac{\gamma-1}{2} M_e^2 \right) = \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \left(\frac{p_{t1} A_{th}}{p_e A_e} \right)^2$$

re-arrange as:

$$\left(M_e^2 \right)^2 + \frac{2}{\gamma-1} M_e^2 - \left(\frac{2}{\gamma-1} \right) \left(\frac{2}{\gamma+1} \right)^{\frac{(\gamma+1)}{\gamma-1}} \left(\frac{p_{t1} A_{th}}{p_e A_e} \right)^2$$

Quadratic equation for M_e^2

- choose + sign to get $M_e^2 > 0$

Gives Mach exit when there is a shock in nozzle.

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$$M_e^2 = \frac{1}{(\gamma - 1)} \left\{ -1 + \left[1 + 2(\gamma - 1) \left(\frac{2}{\gamma + 1} \right)^{\left(\frac{\gamma + 1}{\gamma - 1} \right)} \left(\frac{p_{t1} A_{th}}{p_e A_e} \right)^2 \right]^{\frac{1}{2}} \right\}$$

$\downarrow$ 1.4 $\downarrow$ 0.6 atm $\downarrow$ 0.5

eg. for Me

$$\frac{p_{t1}}{p_e}, \frac{A_{th}}{A_e}$$

known

Using this formula for the current example:

$$M_e = 0.472$$

isentropic tables:

$$\frac{p_{t1}}{p_e} = 1.165 \quad \frac{A_e}{A_e^*} = 1.398$$

$$\frac{p_{t1}}{p_{t2}} = \frac{p_{t1} p_e}{p_e p_{t1}} = (1.165)(0.6) = 0.699$$

for $\frac{p_{t2}}{p_{t1}} = 0.699$

Normal shock tables:

$$M_1 = 2.047 \quad M_2 = 0.5696 \quad \frac{p_2}{p_1} = 4.721$$

Isentropic tables for

$$M_2 = 0.5696 \rightarrow$$

$$\frac{A_2}{A_2^*} = 1.227$$

$$\frac{A_2}{A_{th}} = \frac{A_2}{A_2^*} \frac{A_2^*}{A_e} \frac{A_e}{A_{th}} = (1.227) \left(\frac{1}{1.398} \right) \left(\frac{0.5}{0.25} \right) = 1.755$$

(shock location)

(given)

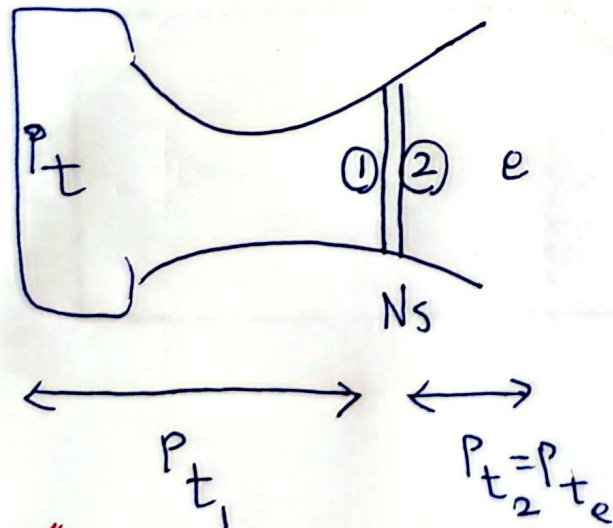
How to solve for shock in nozzle location?

① check if $\frac{P}{P_t} < \frac{P}{P_t}_{crit} < \frac{P}{P_t}_{sub}$ → then shock is in nozzle!

② Use M_e Eq², find M_e ✓

③ with M_e , go to isentropic calc, $\left\{ \begin{array}{l} \text{find } \frac{P_{te}}{P_e} = \frac{P_{t2}}{P_e} = \frac{1}{0.85} \\ \text{find } \frac{A_e}{A^*} = 1.397 \end{array} \right.$

④ Compute = $\frac{P_{te}}{P_{t1}} = \frac{\frac{P_{te}}{P_e}}{\frac{P_e}{P_{t1}}} = 0.699$



"total pressure changes across shock"

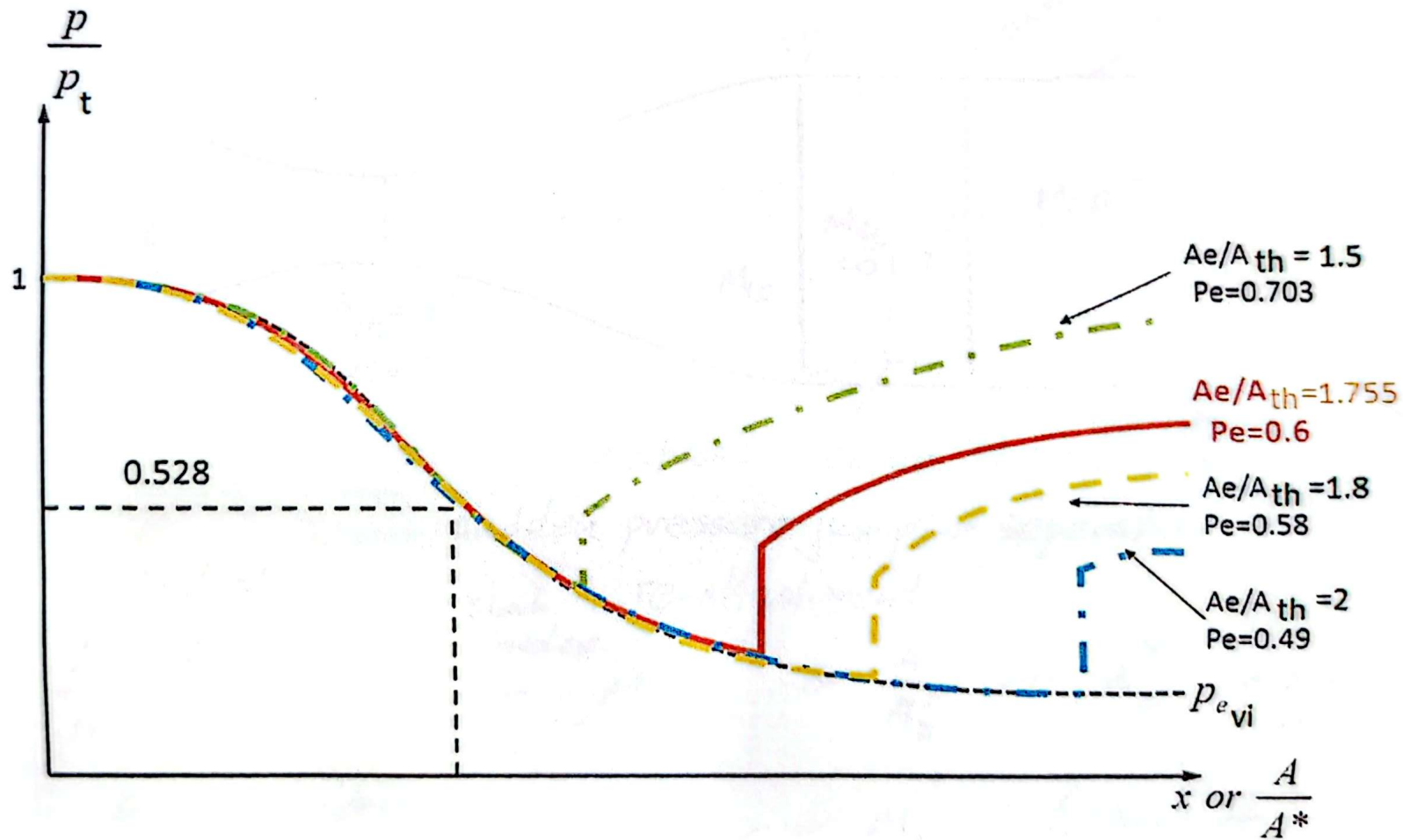
⑤ Go to shock calculator with $\frac{P_{te}}{P_e}$, read M_{shock}

$M_1 = 2.046$

⑥ Go to isentropic table with M_{shock} read $\frac{A}{A^*}$ ✓

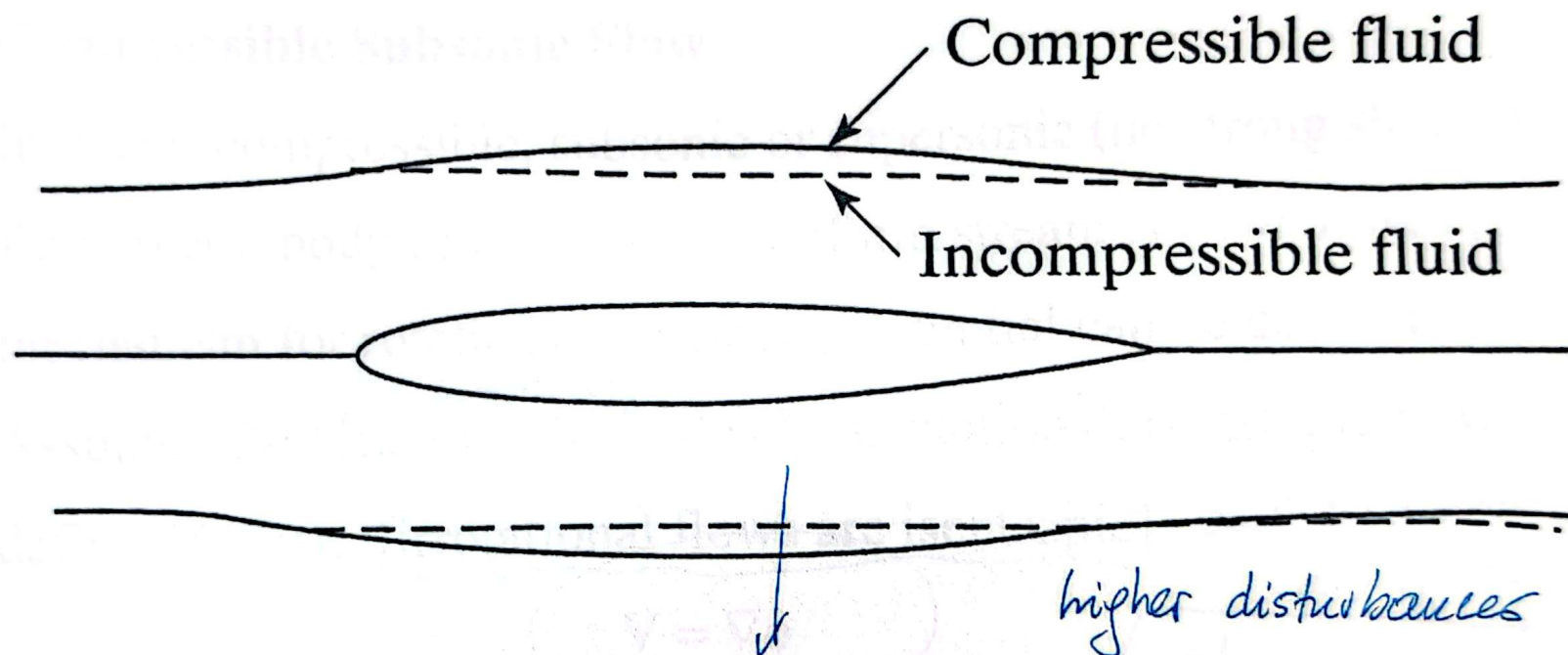
$\frac{A}{A^*} = 1.755$

shock locations ; stronger shocks near the exit



Subsonic Compressible & Transonic Flows [chapter 9]

- Study impact of compressibility to governing equations
- Transform incompressible experimental data or geometry to subsonic compressible Mach numbers. (compressibility "corrections")
- Introduction to transonic flow and its effects.
- Study critical and drag divergence Mach number.
- Introduction to supercritical airfoils, transonic area rule, wing sweep.



A comparison between compressible and incompressible flow shows compressible streamlines opening more and disturbances propagate to ^a larger distance.

Larger speeds lead to critical Mach number ³ and transonic flows

Compressible Subsonic Flow

Inviscid, compressible, subsonic or supersonic (no strong shocks), flow over a body immersed in a uniform stream is *Irrotational* (no mechanism for rotation) $\Rightarrow$ velocity potential can be defined

Assume: 2-D ^{-for simplicity} [text has 3D], steady, irrotational, isentropic flow

define $\phi(x, y)$: [irrotational flows are isentropic] and isentropic flows are irrotational

$$\mathbf{V} = \nabla \phi$$

$$\Rightarrow u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}$$

$$\nabla^2 \phi = 0$$

"potential Eq"
for incomp. flow"

potential flow approximation OK for weak shocks

e. g. if $M_1 = 1 + \epsilon$, ϵ is small can prove that $s_2 - s_1 = O(\epsilon)^3$ (small)
across a shock

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \quad \Rightarrow \quad \begin{array}{l} \text{steady 2-D} \\ \text{continuity equation} \end{array} \quad (9.1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (9.2a)$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y} \quad (9.2b)$$

*x - direction
conservation of
momentum
y - direction*

$$a^2 = \left(\frac{\partial p}{\partial \rho} \right)_s \quad (8.28) \quad (9.3)$$

Combining (9.1)-(9.3) for isentropic flow

$$\left(1 - \frac{u^2}{a^2} \right) \frac{\partial u}{\partial x} + \left(1 - \frac{v^2}{a^2} \right) \frac{\partial v}{\partial y} - \frac{2uv}{a^2} \frac{\partial u}{\partial y} = 0 \quad (9.4a)$$

"Potential Eqⁿ for comp. flow"

$$\phi_x = \frac{\partial \phi}{\partial x}$$

$$\phi_y = \frac{\partial \phi}{\partial y}$$

1 non-linear eq.

$$\left(1 - \frac{\phi_x^2}{a^2}\right) \phi_{xx} + \left(1 - \frac{\phi_y^2}{a^2}\right) \phi_{yy} - \frac{\phi_x \phi_y}{a^2} \phi_{xy} = 0$$

"Non-linear" (9.4b)

full velocity potential equation (only ϕ and a appears)

a can be expressed in terms of ϕ :

$$(\phi_x = \frac{\partial \phi}{\partial x}, \phi_y = \frac{\partial \phi}{\partial y})$$

$$\frac{a^2}{a_\infty^2} = 1 - \frac{\gamma - 1}{2} \left(\frac{u^2 + v^2}{U_\infty^2} - 1 \right) M_\infty^2 \Rightarrow \frac{a^2}{a_\infty^2} = 1 - \frac{\gamma - 1}{2} \left(\frac{\phi_x^2 + \phi_y^2}{U_\infty^2} - 1 \right) M_\infty^2 \quad (9.6)$$

Eq. (9.4) represents continuity momentum & energy can be easily

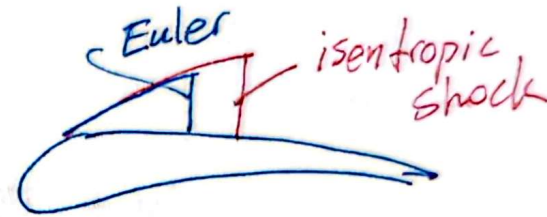
(CFD!) solved (only 1 equation); find ϕ then find u, v, a ; $M = \frac{U}{a}$; $T,$

p, ρ from isentropic flow relations (isentropic tables) can be used to

represent weak shocks (transonic flow)

- not very strong shocks ($\Rightarrow$ isentropic)

[isentropic shocks are stronger]



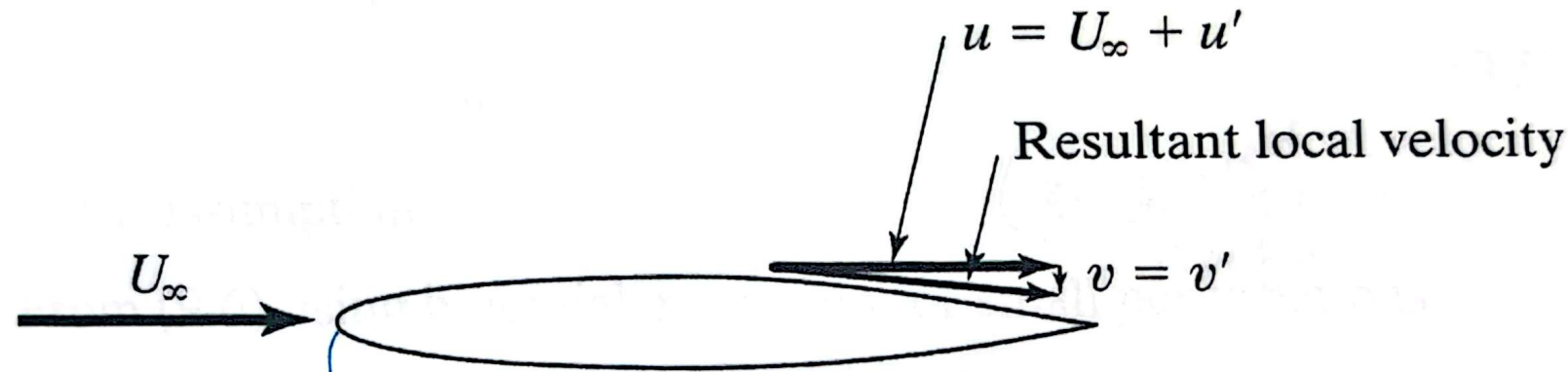
- entropy corrections can be added to the model.

Next level-up Euler eqs. (more expensive)

Still eq. (5.12)^{9.4} is nonlinear, solve computationally (finite difference methods) $\rightarrow$ used a lot in industry (e.g. flow around transonic wings)

linear approximations (subsonic – supersonic)

The linearized velocity potential equation



Let u' and v' be perturbations of the velocity in the x and y directions

Perturbation velocities u' and v' can be large near the leading edge

stagnation region.

$$|u| \sim |v_\infty|$$

$$u = U_\infty + u'; \quad v = v'$$

C_p might not be accurate near (10%) leading edge

(similar to thin airfoil theory)

-but lift is ok!

From (9.4a) for small perturbations

$$\left(1 - \frac{u^2}{a^2}\right) \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (9.5)$$

Extra assumptions:

*(go fast over
- algebra 9.6 - 9.10)*

From (9.6) using binomial expansion for small perturbations

$$\frac{a_\infty^2}{a^2} = 1 + \frac{\gamma - 1}{2} M_\infty^2 \left(2 \frac{u'}{U_\infty} + \frac{u'^2 + v'^2}{U_\infty^2} \right) \quad (9.7)$$

To simplify (9.5): *note that*

$$1 - \frac{u^2}{a^2} = 1 - \frac{U_\infty^2 + 2u'U_\infty + u'^2}{U_\infty^2} M_\infty^2 \frac{a_\infty^2}{a^2} \quad (9.8)$$

($u = U_\infty + u'$)

Substituting (9.7) to (9.8) and neglecting high-order terms

$$1 - \frac{u^2}{a^2} = 1 - M_\infty^2 \left[1 + \frac{2u'}{U_\infty} \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) \right] \quad (9.9)$$

Substitute (9.9) to (9.5)

$$(1 - M_\infty^2) \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = M_\infty^2 \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) \frac{2u'}{U_\infty} \frac{\partial u}{\partial x}$$

Using disturbances:

$$(1 - M_\infty^2) \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} = \frac{2}{U_\infty} \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) M_\infty^2 u' \frac{\partial u'}{\partial x} \quad (9.10)$$

Right hand side can be neglected if *disturbances are small*

$$\Rightarrow (1 - M_\infty^2) \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} = 0 \quad (9.11)$$

Perturbation potential ϕ' :

$$\phi = U_\infty x + \phi'$$

$$\Rightarrow \frac{\partial \phi}{\partial x} = U_\infty + \frac{\partial \phi'}{\partial x}, \quad \frac{\partial \phi}{\partial y} = \frac{\partial \phi'}{\partial y}$$

$$(1 - M_\infty^2) \phi_{xx} + \phi_{yy} = 0 \Rightarrow \beta^2 \phi_{xx} + \phi_{yy} = 0 \quad (9.13)$$

where $\beta \equiv \sqrt{1 - M_\infty^2}$

linearized potential flow

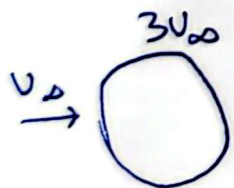
$$\beta = \sqrt{1 - M_\infty^2}$$

Prandtl Glauert equation

$$\phi_{xx} = \frac{\partial^2 \phi}{\partial x^2}$$

$$\phi_{yy} = \frac{\partial^2 \phi}{\partial y^2}$$

$$\beta^2 \phi_{xx} + \phi_{yy} = 0$$



Not small disturbance



Small disturbance

Assumptions:

- 1) small perturbations (thin bodies, small α)
- 2) subsonic & supersonic M_∞

$$\beta^2 \phi_{xx} + \phi_{yy} = 0$$

Not good for: thick bodies or large α

transonic flow $0.8 < M_\infty < 1.2$

hypersonic flow $M_\infty > 5$

hierarchy:

Navier Stokes; Euler; full potential; TSD; linear potential



our Eq: $\beta^2 \phi_{xx} + \phi_{yy} = 0$ $\xrightarrow{\quad}$ $\phi_{x'x'} + \phi_{y'y'} = 0$

The Gothert Transformation

Objective: transform (9.13) to Laplace's equation:

$$x' = \frac{x}{\sqrt{1-M_\infty^2}} = \frac{x}{\beta} \quad y' = y \quad (9.14)$$

Then (9.13) becomes: $\phi_{x'x'} + \phi_{y'y'} = 0$ (9.15)

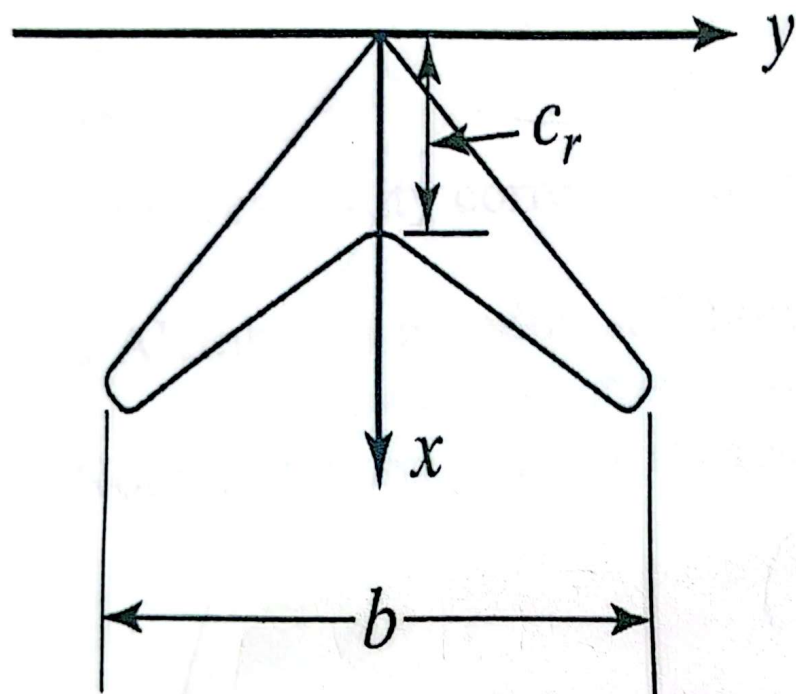
Disturbance pressure coefficient (linear approximation - similar to thin airfoil theory) $C_p = -\frac{2u'}{U_\infty} = -\frac{2\phi'_{x'}}{U_\infty}$ (9.16a)
depends only on u' (not v')

Therefore:

$$C_p = +\frac{C'_p}{\beta} \quad (9.16b)$$

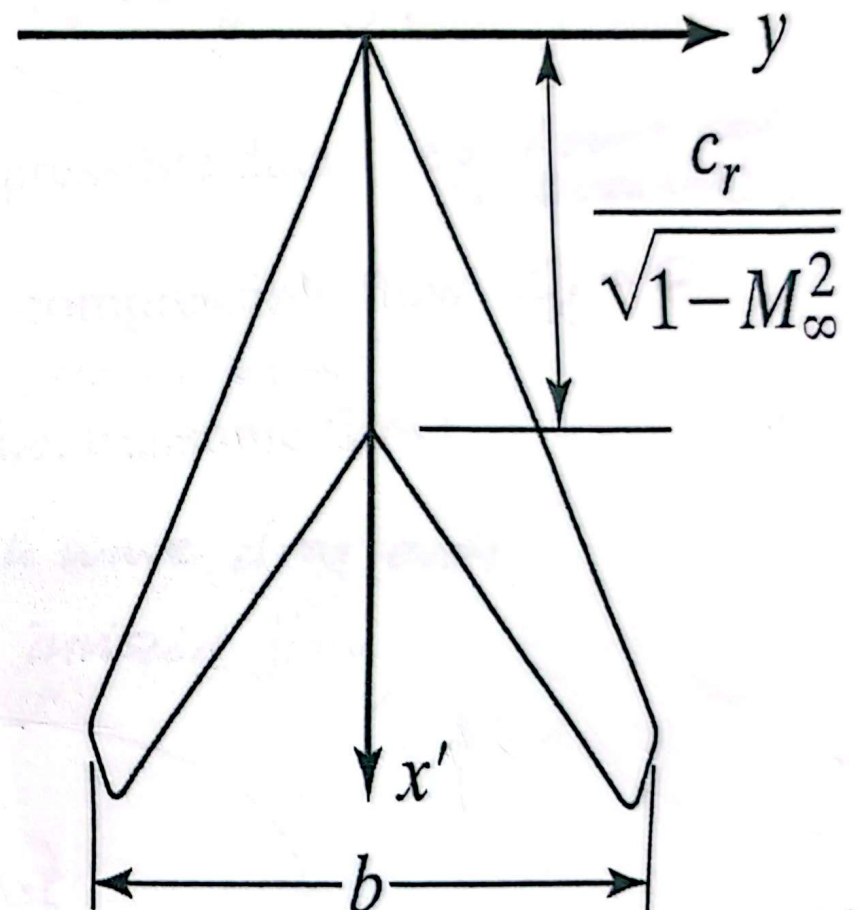
Prandtl-Glauert/compressibility correction

compressible $\rightarrow$ incompressible



compressible flow

(a)



(b)

incompressible flow

Also:
(integrating C_p)

$$C_l = + \frac{C'_l}{\beta}$$

$$C_m = + \frac{C'_m}{\beta}$$

Compressibility correction to incompressible data (e.g. Abbott and Doenhof)

d'Alembert's paradox still holds for compressible flow $C_d = 0$
(potential flow cannot predict drag)

Not valid when a shock is present (i.e. transonic flow)

(shock wave drag even for inviscid flow)

$$C_{e, \text{comp}} = \frac{2\pi(\alpha - \alpha_{l_0})}{\beta}$$

Sub

Example

the low-speed for NACA 2412 at $\alpha = 4^\circ$ has $C'_\ell = 0.65$ (incompressible)

find C_ℓ at $M_\infty = 0.7$ using Prandtl-Glauert Rule

$$C_\ell = \frac{C'_\ell}{\beta} = \frac{C'_\ell}{\sqrt{1 - M_\infty^2}} = \frac{0.65}{\sqrt{1 - (0.7)^2}} = 0.910$$

For a thin airfoil: $C_\ell = 2\pi\alpha$
 $\swarrow$
 sym.

Using the Prandtl-Glauert Rule: $C_\ell = \frac{2\pi\alpha}{\beta}$

note: $M_\infty = 0.7 \rightarrow \beta = \sqrt{1 - M_\infty^2} = 0.714$

$$C_\ell = C'_\ell / \beta \quad \text{error} \approx 5\%$$

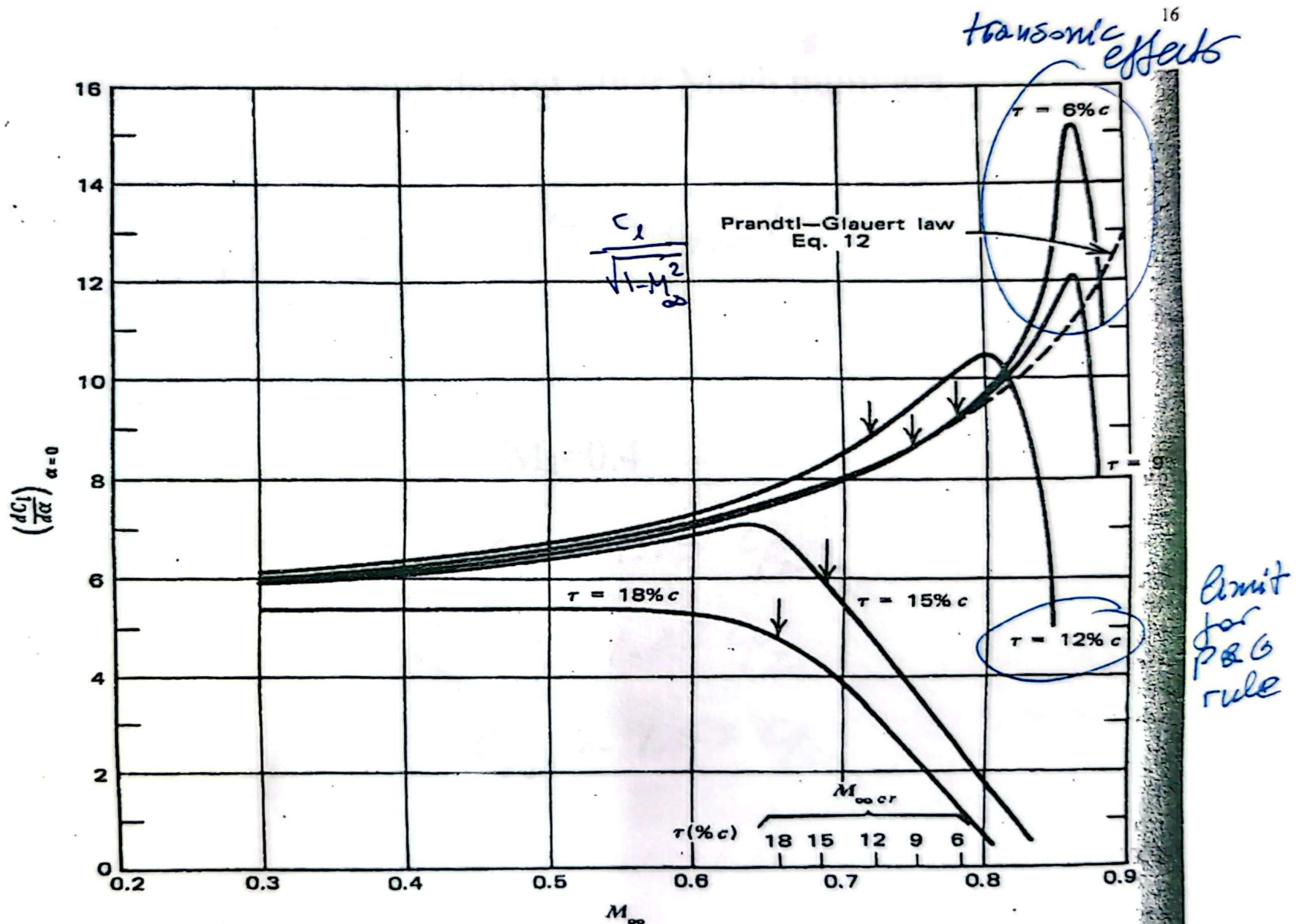


Fig. 4. Measured $dc_l/d\alpha$ versus M_∞ for symmetric airfoils of different maximum thickness and comparison with Prandtl-Glauert law for $dc_l/d\alpha = 6.0$ per rad at $M_\infty = 0.3$; $M_{\infty cr}$ each curve is designated by an arrow and by a point on the abscissa. (Mair and Beavan, 1965: Howarth, 1953. Courtesy of Oxford University Press)

$$M_2 = 0.4$$

$$M_2$$

Given data at $M=0.4$ scale data at other Mach numbers

$$\frac{C_{p1}}{C_{p2}} = \frac{C_p' / \sqrt{1-M_{1\infty}^2}}{C_p' / \sqrt{1-M_{2\infty}^2}} \Rightarrow C_{p1} = C_{p2} \sqrt{\frac{1-M_{2\infty}^2}{1-M_{1\infty}^2}}$$

e.g.

$$M_1 =$$

$$M_2 = 0.4$$

$$0.6$$

$$C_{p1} = 1.15 C_{p2}$$

$$0.7$$

$$C_{p1} = 1.28 C_{p2}$$

$$0.8$$

$$C_{p1} = 1.53 C_{p2}$$

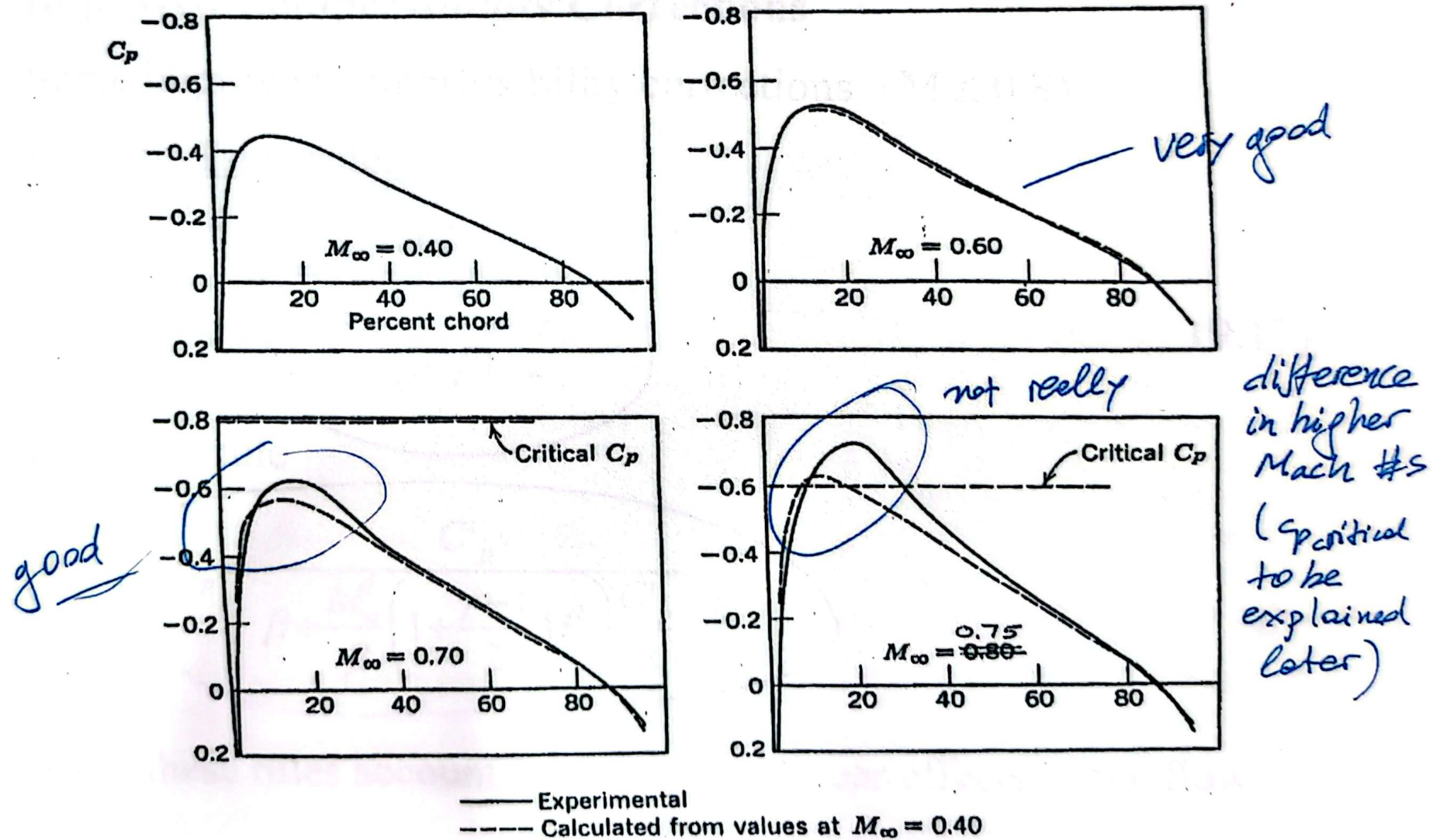


FIG. 10-1 Comparison of Prandtl-Glauert similarity rule with experiment. (Experimental data for NACA 0012 airfoil, taken from *NACA Tech. Note 2174* by J. L. Amick.)

Improved Compressibility Corrections

Some improved compressibility corrections ($M \leq 0.8$)

Karman-Tsien rule:

$$C_p = \frac{C'_p}{\beta + \frac{M_\infty^2}{(1+\beta)} \frac{C'_p}{2}} \quad (9.17)$$

Laitone's rule

$$C_p = \frac{C'_p}{\beta + \frac{M_\infty^2}{\beta} \left(1 + \frac{\gamma-1}{2} M_\infty^2 \right) \frac{C'_p}{2}} \quad (9.18)$$

Both these rules account for some non-linear effects of the flow

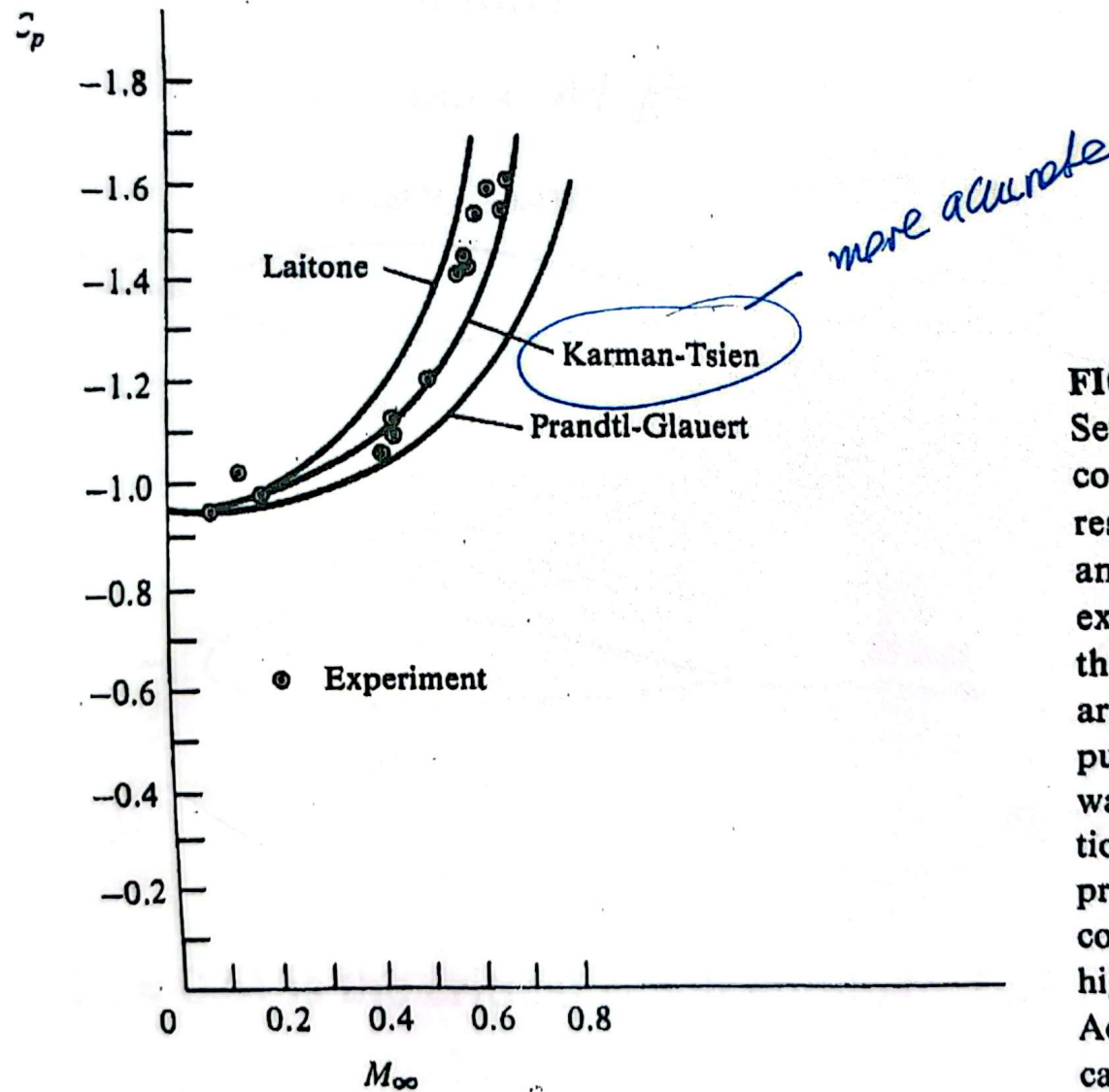
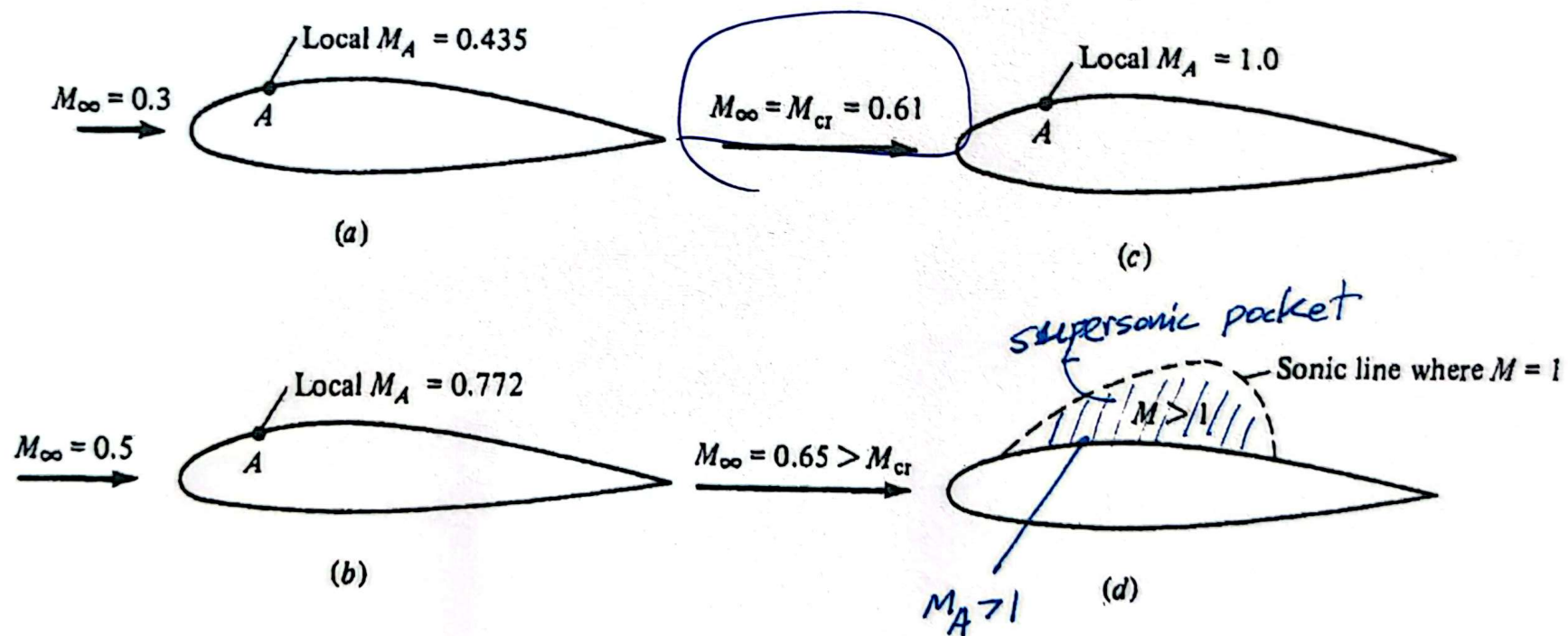


FIGURE 11.4

Several compressibility corrections compared with experimental results for an NACA 4412 airfoil at an angle of attack $\alpha = 1^\circ 53'$. The experimental data are chosen for their historical significance; they are from NACA report no. 646, published in 1938 (Ref. 30). This was the first major NACA publication to address the compressibility problem in a systematic fashion; it covered work performed in the 2-ft high-speed tunnel at the Langley Aeronautical Laboratory and was carried out during 1935-1936.

Critical Mach number

$A = \max M \#$



$M_\infty = 0.61$ is the critical Mach number (~~set~~ Max velocity = $M = 1$)